24.

Trigonometry

Type - 4

1. If tan A + cot A = 2, then find the value of $\tan^2 A + \cot^2 A$

(a) 4 (b) 2
(c) 1 (d) 1/2

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans : (b) Given, tan A + cot A = 2
Square of both sides
$(\tan A + \cot A)^2 = 4$
$\tan^2 A + \cot^2 A + 2\tan A \cot A = 4$
$\tan^2 A + \cot^2 A + 2\times1 = 4$ ($\because \tan A \cot A = 1$)
$\tan^2 A + \cot^2 A = 4 - 2 = 2$

2. If $\cos^2 x + \sin x = 5/4$, then find the value 'sin x'

(a) 3/4 (b) 1/2
(c) - 1/2 (d) 3/2

RRB JE - 24/05/2019 (Shift-I)

Ans : (b)

$\because \cos^2 x + \sin x = \frac{5}{4}$

$1 - \sin^2 x + \sin x = \frac{5}{4}$

$\sin^2 x - \sin x + \frac{1}{4} = 0$

$\sin^2 x - 2\times\frac{1}{2}\times\sin x + \frac{1}{4} = 0$

$\left(\sin x - \frac{1}{2}\right)^2 = 0$

$\sin x - \frac{1}{2} = 0$

$\sin x = \frac{1}{2}$

3. If cosecθ – sinθ = P and secθ – cosθ = q, then which of the following is correct?

(a) $(p^2q)^{\frac{2}{3}} - (pq^2)^{\frac{2}{3}} = 1$

(b) $\sin\theta\sec\theta = \frac{1}{P}$

(c) $\sin\theta\tan\theta = \frac{1}{q}$

(d) $(p^2q)^{\frac{2}{3}} + (pq^2)^{\frac{2}{3}} = 1$

RRB RPF SI – 16/01/2019 (Shift-III)

Ans. (d) Given–

$\operatorname{cosec}\theta - \sin\theta = p$ $\qquad$ $\sec\theta - \cos\theta = q$

$\therefore \frac{1}{\sin\theta} - \sin\theta = p$ $\qquad$ $\frac{1}{\cos\theta} - \cos\theta = q$

$\frac{1-\sin^2\theta}{\sin\theta} = p$ $\qquad$ $\frac{1-\cos^2\theta}{\cos\theta} = q$

$\cos^2\theta = p\sin\theta$ $\qquad$ $\sin^2\theta = q\cos\theta$

$p = \frac{\cos^2\theta}{\sin\theta}$ $\qquad$ $q = \frac{\sin^2\theta}{\cos\theta}$

$p^2q = \frac{\cos^4\theta}{\sin^2\theta}\times\frac{\sin^2\theta}{\cos\theta}$

$p^2q = \cos^3\theta$

$(p^2q)^{\frac{1}{3}} = \cos\theta$

$(p^2q)^{\frac{2}{3}} = \cos^2\theta$....(i)

Similar

$(q^2p)^{\frac{2}{3}} = \sin^2\theta$.....(ii)

Adding equation (i) equation (ii)

$(p^2q)^{\frac{2}{3}} + (pq^2)^{\frac{2}{3}} = 1$ ($\because \sin^2\theta + \cos^2\theta = 1$)

4. If secθ + tanθ = 12.5, then secθ - tanθ = ?

(a) 1 (b) 0.08
(c) 0.8 (d) 0.4

RRB RPF Constable – 19/01/2019 (Shift-II)

Ans. (b) : $\sec\theta + \tan\theta = 12.5$
Multiplying to $(\sec\theta - \tan\theta)$ in both sides

$\operatorname{Sec}^2\theta - \tan^2\theta = 12.5(\sec\theta - \tan\theta)$ $\quad (\because \sec^2 - \tan^2\theta = 1)$

$1 = 12.5(\sec\theta - \tan\theta)$

or, $\sec\theta - \tan\theta = \frac{1}{12.5}$

$\boxed{\operatorname{Sec}\theta - \tan\theta = 0.08}$

5. If $\sec^4\theta - \sec^2\theta = 3$, then $\tan^4\theta + \tan^2\theta = ?$

(a) 2 (b) 0
(c) 3 (d) 1

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (c) If $\operatorname{Sec}^4\theta - \operatorname{Sec}^2\theta = 3$

$\operatorname{Sec}^2\theta(\operatorname{Sec}^2\theta - 1) = 3$ $\quad [\because \sec^2\theta - \tan^2\theta = 1]$

$(\tan^2\theta + 1)\tan^2\theta = 3$

$\tan^4\theta + \tan^2\theta = 3$

So $\boxed{\tan^4\theta + \tan^2\theta = 3}$

6. If $\cot^4\theta+\cot^2\theta = 2.2$, then $\text{cosec}^4\theta - \text{cosec}^2\theta = ?$

(a) 0 (b) 1.1
(c) 2.2 (d) 3.3

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (c) $\cot^4\theta + \cot^2\theta = 2.2$
$\cot^2\theta (\cot^2\theta + 1) = 2.2$ $\boxed{\therefore \cot^2\theta = \text{cosec}^2\theta - 1}$
$(\text{cosec}^2\theta - 1) (\text{cosec}^2\theta - 1 + 1) = 2.2$
$(\text{cosec}^2\theta - 1) (\text{cosec}^2\theta) = 2.2$
$\text{cosec}^4\theta - \text{cosec}^2\theta = 2.2$

7. If $(\tan\theta + \cot\theta) = 5$, then find the value of $(\tan^2\theta + \cot^2\theta)$?

(a) 23 (b) 27
(c) 25 (d) 21

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (a) $\tan\theta + \cot\theta = 5$ ----------(i)
Square of both sides
$\tan^2\theta+\cot^2\theta+2\tan\theta.\cot\theta = 25$ $[\tan\theta.\cot\theta=1]$
$\tan^2\theta+\cot^2\theta = 25-2 = 23$

8. If $\sec\theta + \tan\theta = 3.2$, then what is $\sec\theta$?

(a) 2.28 (b) 1.6
(c) 1.75625 (d) 1.92625

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (c) $\sec\theta + \tan\theta = 3.2 = \frac{32}{10} = \frac{16}{5}$.....(i)
multiplying to $(\sec\theta - \tan\theta)$ of equation (i) in both sides.
$(\sec\theta + \tan\theta) (\sec\theta - \tan\theta) = \frac{16}{5} (\sec\theta - \tan\theta)$
$\therefore$ $[\sec^2\theta - \tan^2\theta = 1]$
$\sec\theta - \tan\theta = \frac{5}{16}$.....(ii)
from equation (i) + (ii)
$2\sec\theta = \frac{16}{5} + \frac{5}{16} = \frac{281}{80} \Rightarrow 2\sec\theta = 3.5125$
$\sec\theta = \frac{3.5125}{2} = 1.75625$

9. If $\sin\theta - \cos\theta = 0$, than what is the value of $\sin^4\theta + \cos^4\theta + \tan^4\theta$

(a) $\frac{5}{4}$ (b) $\frac{3}{2}$ (c) $\frac{7}{4}$ (d) 2

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (b) : $\sin\theta - \cos\theta = 0$ then $\sin^4\theta + \cos^4\theta + \tan^4\theta$ value will be–
$\sin\theta - \cos\theta = 0$
$\sin\theta = \cos\theta$
$\because$ $\theta = 45°$ on $\sin\theta = \cos\theta$
$\theta = 45°$ putting
$\sin^4\theta + \cos^4\theta + \tan^4\theta$
$= \sin^4(45°) + \cos^4(45°) + \tan^4(45°)$
$= \left(\frac{1}{\sqrt{2}}\right)^4 + \left(\frac{1}{\sqrt{2}}\right)^4 + (1)^4$
$= \frac{1}{4} + \frac{1}{4} + 1 = \frac{2}{4} + 1 = \frac{1}{2} + 1 = \frac{3}{2}$

10. If $\cot^4\theta + \cot^2\theta = 3.6$, then $\text{cosec}^4\theta - \text{cosec}^2\theta = ?$

(a) 0.6 (b) 3.6
(c) 2.4 (d) 1.8

RRB Group-D – 17/09/2018 (Shift-III)

Ans. (b) : $\cot^4\theta + \cot^2\theta = 3.6$
$\cot^2\theta (1 + \cot^2\theta) = 3.6$
$$\begin{cases} \text{cosec}^2\theta - \cot^2\theta = 1 \\ \text{cosec}^2\theta = 1 + \cot^2\theta \\ \text{cosec}^2\theta - 1 = \cot^2\theta \end{cases}$$
$\boxed{\cot^2\theta.\ \text{cosec}^2\theta = 3.6}$ .....(i)
$(\text{cosec}^2\theta - 1).\ \text{cosec}^2\theta = 3.6$
$\text{cosec}^4\theta - \text{cosec}^2\theta = 3.6$

11. If $\cos x + \sin x = \sqrt{2}\cos x$, then what is the value of $\cot x$?

(a) $\sqrt{2}$ (b) 1
(c) $\sqrt{2}+1$ (d) $\sqrt{2}-1$

RRB Group-D – 25/09/2018 (Shift-I)

Ans : (c)
$\cos x + \sin x = \sqrt{2}\cos x$
sinx divide from sinx,
$\cot x + 1 = \sqrt{2}\cot x$
$\sqrt{2}\cot x - \cot x = 1$
$\cot x\left(\sqrt{2}-1\right) = 1$
$\cot x = \frac{1}{\sqrt{2}-1}$
or
$$\cot x = \frac{\left(\sqrt{2}+1\right)}{\left(\sqrt{2}-1\right)\left(\sqrt{2}+1\right)}$$
$$\cot x = \frac{\sqrt{2}+1}{2-1}$$
$$\cot x = \sqrt{2}+1$$

12. If $\sec\theta + \tan\theta = 1.25$, then $\sec\theta - \tan\theta = ?$

(a) 1 (b) 0.80
(c) 0.75 (d) 0.25

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (b) $\sec\theta + \tan\theta = 1.25 = \frac{125}{100}$
$$\frac{(\sec\theta + \tan\theta)(\sec\theta - \tan\theta)}{(\sec\theta - \tan\theta)} = \frac{125}{100}$$
$$\frac{\sec^2\theta - \tan^2\theta}{(\sec\theta - \tan\theta)} = \frac{125}{100} \qquad \left\{\sec^2\theta - \tan^2\theta = 1\right\}$$
$$\frac{1}{(\sec\theta - \tan\theta)} = \frac{125}{100}$$
$$\sec\theta - \tan\theta = \frac{100}{125}$$
$$\sec\theta - \tan\theta = 0.80$$

13. If $\tan^4\theta + \tan^2\theta = 11$, then $\sec^4\theta - \sec^2\theta = ?$

(a) 12 (b) 11
(c) 13 (d) 10

RRB Group-D – 27/09/2018 (Shift-III)

Ans : (b) $\because \tan^4\theta + \tan^2\theta = 11$
$\therefore \tan^2\theta\,(1 + \tan^2\theta) = 11$
$\tan^2\theta.\ \sec^2\theta = 11$
$(\sec^2\theta - 1)\sec^2\theta = 11$ $\{\because 1 + \tan^2\theta = \sec^2\theta\}$
$\sec^4\theta - \sec^2\theta = 11$

14. If $\cot^4\theta + \cot^2\theta = 4$, then what is the value of $\text{cosec}^4\theta - \text{cosec}^2\theta = ?$

(a) 4 (b) 0
(c) 2 (d) 3

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (a) $\cot^4\theta + \cot^2\theta = 4$
$(\text{cosec}^2\theta - 1)^2 + \text{cosec}^2\theta - 1 = 4$
$\text{cosec}^4\theta + 1 - 2\,\text{cosec}^2\theta + \text{cosec}^2\theta - 1 = 4$
$\text{cosec}^4\theta - \text{cosec}^2\theta = 4$

15. If $\sec\theta + \tan\theta = 8$, then $\sec\theta - \tan\theta = ?$

(a) 0.5 (b) 0.625
(c) 0.125 (d) 0.8

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (c) Given– $\sec\theta + \tan\theta = 8$ .....(i)
Multipling to $(\sec\theta - \tan\theta)$ of equation (i) in both sides–
$(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 8(\sec\theta - \tan\theta)$
$\therefore$ $(\sec^2\theta - \tan^2\theta = 1)$
$(\sec\theta - \tan\theta) = \frac{1}{8}$
$(\sec\theta - \tan\theta) = 0.125$

16. If $\tan\alpha = 3 - 2\sqrt{2}$, then what is the value of $\tan\alpha - \cot\alpha$?

(a) -4 (b) $3 + 2\sqrt{2}$
(c) $-4\sqrt{2}$ (d) $-8\sqrt{2}$

RRB Paramedical Exam – 20/07/2018 (Shift-II)

Ans. (c) $\tan\alpha = 3 - 2\sqrt{2}$ ...(i)
then, $\tan\alpha - \cot\alpha = ?$
$= \tan\alpha - \frac{1}{\tan\alpha}$ (i) from equation
$= \left(3 - 2\sqrt{2}\right) - \frac{1}{3 - 2\sqrt{2}}$
$= \left(3 - 2\sqrt{2}\right) - \frac{\left(3 + 2\sqrt{2}\right)}{\left(3 - 2\sqrt{2}\right)\left(3 + 2\sqrt{2}\right)}$
$= \left(3 - 2\sqrt{2}\right) - \frac{\left(3 + 2\sqrt{2}\right)}{9 - 8}$
$= \left(3 - 2\sqrt{2}\right) - \frac{\left(3 + 2\sqrt{2}\right)}{1}$
$= 3 - 2\sqrt{2} - 3 - 2\sqrt{2}$
$= -4\sqrt{2}$

17. If $\sec\theta + \tan\theta = 6$, then $\sec\theta = ?$

(a) $3\frac{1}{12}$ (b) $3\frac{1}{6}$ (c) 3 (d) $3\frac{1}{3}$

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (a) $\sec\theta + \tan\theta = 6$ then $\sec\theta = ?$
$\because$ $\sec^2\theta - \tan^2\theta = 1$
$\therefore$ $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$
$\therefore$ $\sec\theta - \tan\theta = \frac{1}{\sec\theta + \tan\theta}$
$\therefore$ $\sec\theta - \tan\theta = \frac{1}{6}$...(i)
$\sec\theta + \tan\theta = 6$...(ii)
Adding of equation (i) and (ii)
$\sec\theta - \tan\theta = \frac{1}{6}$.....(i)
$\sec\theta + \tan\theta = 6$.....(ii)
$2\sec\theta = \frac{1}{6} + 6$
$2\sec\theta = \frac{37}{6}$
$\sec\theta = \frac{37}{12} = 3\frac{1}{12}$

18. If $\sec\theta + \tan\theta = 2.5$, then what is the value of $\sec\theta - \tan\theta = ?$

(a) 1 (b) 0.4 (c) 0.8 (d) 0.6

RRB Group-D – 02/11/2018 (Shift-I)

Ans. (b) $\sec\theta + \tan\theta = 2.5$
Multipling to $(\sec\theta - \tan\theta)$ in both sides
$(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 2.5(\sec\theta - \tan\theta)$
$\sec^2\theta - \tan^2\theta = 2.5\,(\sec\theta - \tan\theta)$
$\because$ $\boxed{\sec^2\theta - \tan^2\theta = 1}$
$\therefore$ $\frac{1}{2.5} = \sec\theta - \tan\theta$
$\sec\theta - \tan\theta = \frac{10}{25}$
$\boxed{\sec\theta - \tan\theta = 0.4}$

19. If $\cos\theta + \sin\theta = m$, $\sec\theta + \text{cosec}\theta = n$, then what is the value of m/n?

(a) 1 (b) $\sin\theta\cos\theta$
(c) $\sec\theta\,\text{cosec}\,\theta$ (d) $\cot\theta\tan\theta$

RRB NTPC 03.04.2016 Shift : 1

Ans : (b) $\cos\theta + \sin\theta = m$ .....(i)
and $\sec\theta + \text{cosec}\,\theta = n$
$\frac{1}{\cos\theta} + \frac{1}{\sin\theta} = n$
$\Rightarrow$ $\frac{\sin\theta + \cos\theta}{\cos\theta\sin\theta} = n$
$\Rightarrow$ $\frac{m}{\cos\theta\sin\theta} = n$ [$\sin\theta + \cos\theta = m$ from equation (i)]
$\Rightarrow$ $\frac{m}{n} = \sin\theta\cos\theta$

20. If $\sec\theta - \tan\theta = \frac{1}{3}$, then what is the value of $\sec\theta + \tan\theta$?

(a) 1 (b) 9 (c) 3 (d) 1/9

RRB NTPC 28.03.2016 Shift : 2

Ans : (c) $\text{sce}\theta - \tan\theta = \frac{1}{3}$

$\because \sec^2\theta - \tan^2\theta = 1$

$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$

$\frac{1}{3}(\sec\theta + \tan\theta) = 1$

$\sec\theta + \tan\theta = 3$

21. If $4\sin\theta - 3\cos\theta = 0$, then $\sec\theta\text{cosec}\theta$ = ?

(a) 5/12 (b) 25/12

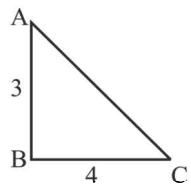
(c) 13/12 (d) 12/5

RRB NTPC 29.04.2016 Shift : 2

Ans : (b) Given—

$4\sin\theta - 3\cos\theta = 0$

$4\sin\theta = 3\cos\theta$



$\frac{\sin\theta}{\cos\theta} = \frac{3}{4}$

$\tan\theta = \frac{3}{4} = \frac{\text{Perpendicular}}{\text{base}}$

from the Pythagoras theorem-

$(\text{hypotenuse})^2 = (\text{Perpendicular})^2 + (\text{base})^2$

$(AC)^2 = 3^2 + 4^2 = 9 + 16 = 25$

AC (hypotenuse) = $\sqrt{25} = 5$

Then, $\sec\theta.\ \text{cosec}\theta = \frac{\text{hypotenuse}}{\text{base}} \times \frac{\text{hypotenuse}}{\text{perpendicular}}$

$= \frac{5}{4} \times \frac{5}{3}$

$\sec\theta.\ \text{cosec}\theta = \frac{25}{12}$

22. If $\sin x + \cos x = \sqrt{2}\sin x$, then the value of tan x is:

(a) $\sqrt{2}+1$ (b) 1

(c) $\sqrt{2}-1$ (d) $\sqrt{2}$

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (a) $\sin x + \cos x = \sqrt{2}\sin x$

divide by cosx in both sides

$\frac{\sin x}{\cos x} + \frac{\cos x}{\cos x} = \frac{\sqrt{2}\sin x}{\cos x}$

$\tan x + 1 = \sqrt{2}\tan x$

$1 = \sqrt{2}\tan x - \tan x$

$1 = \tan x\left(\sqrt{2}-1\right)$

$\tan x = \frac{1}{\sqrt{2}-1}$

$\tan x = \frac{\sqrt{2}+1}{(\sqrt{2}-1)(\sqrt{2}+1)} = \sqrt{2}+1$

23. If $\sec\theta + \tan\theta = 4$, $\sec\theta - \tan\theta$ = ?

(a) 1 (b) 0.75

(c) 0.25 (d) 0.5

RRB ALP & Tec. (13-08-18 Shift-II)

Ans : (c) Given that,

$(\sec\theta + \tan\theta) = 4$

$\Rightarrow (\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 4(\sec\theta - \tan\theta)$

$\Rightarrow (\sec^2\theta - \tan^2\theta) = 4(\sec\theta - \tan\theta)$

$\Rightarrow 1 = 4(\sec\theta - \tan\theta)$

$\Rightarrow (\sec\theta - \tan\theta) = 1/4$

$\Rightarrow (\sec\theta - \tan\theta) = 0.25$

24. If $\text{cosec}\,\theta + \cot\theta = 2$, then $\cot\theta$ = ?

(a) 0 (b) 1

(c) 0.75 (d) 0.5

RRB ALP & Tec. (10-08-18 Shift-I)

Ans : (c) given- $(\text{cosec}\theta + \cot\theta = 2)$

and $\text{cosec}\theta + \cot\theta = 2$ ..... (i)

$\text{cosec}^2\theta - \cot^2\theta = 1$ (from formula)

$\because (\text{cosec}\,\theta + \cot\theta)(\text{cosec}\,\theta - \cot\theta) = 1$

$(2)(\text{cosec}\theta - \text{tot}\theta) = 1$

$\therefore \text{cosec}\theta - \cot\theta = \frac{1}{2}$ ..... (ii)

Subtracting equation (ii) from (i)

$2\cot\theta = 2 - \frac{1}{2} = 1.50$

$\therefore \cot\theta = \frac{1.50}{2} = 0.75$

Type - 2

25. What is the value of $\sqrt{\cot^2\theta - \cos^2\theta}$

(a) 1 (b) $\cot\theta\cos ec\theta$

(c) $\cot\theta\cos\theta$ (d) $\cos\theta\cos ec\theta$

RRB JE - 23/05/2019 (Shift-I)

Ans : (c)

$\sqrt{\cot^2\theta - \cos^2\theta} = \sqrt{\frac{\cos^2\theta}{\sin^2\theta} - \cos^2\theta}$

$= \sqrt{\frac{\cos^2\theta - \sin^2\theta\cos^2\theta}{\sin^2\theta}}$

$= \sqrt{\frac{\cos^2\theta\left(1-\sin^2\theta\right)}{\sin^2\theta}} = \sqrt{\cot^2\theta.\left(1-\sin^2\theta\right)} = \cot\theta.\ \cos\theta$

26. What is the value of $\cos^4 A - \sin^4 A$

(a) cos2A (b) $\sin^2 A + \cos^2 A$

(c) 0 (d) 2

RRB JE - 23/05/2019 (Shift-I)

Ans : (a) $\cos^4 A - \sin^4 A = (\cos^2 A)^2 - (\sin^2 A)^2$

$= (\cos^2 A + \sin^2 A)(\cos^2 A - \sin^2 A)$

$= \cos^2 A - \sin^2 A$ $\{\because \cos^2 A + \sin^2 A = 1\}$

$= \cos 2A$

27. What is the value of $(1 - \cos^2\theta)(\cot^2\theta + 1) - 1$

(a) 0 (b) $\sec^2\theta$

(c) 2 (d) -2

RRB JE - 23/05/2019 (Shift-I)

Ans : (a) $(1-\cos^2\theta)(\cot^2\theta+1)-1$

$\{1-\cos^2\theta=\sin^2\theta$

$1+\cot^2\theta=\text{cosec}^2\theta\}$

$=\sin^2\theta\times\text{cosec}^2\theta-1=1-1=0$

28. Simplify : cos θ/(1 + sin θ)

(a) cosec θ + cot θ (b) sec θ - tan θ

(c) cosec θ - cot θ (d) sec θ + tan θ

RRB RPF Constable – 18/01/2019 (Shift-III)

Ans : (b) $\frac{\cos\theta}{1+\sin\theta}$

Multiplying to (1 – sin θ) in denominator and numerator

$=\frac{(1-\sin\theta)\times\cos\theta}{\cos^2\theta}=(\sec\theta-\tan\theta)$

29. Find the value of 'θ'

$\frac{\cos\theta}{1-\sin\theta}-\frac{\cos\theta}{1+\sin\theta}=2$

(a) 45 (b) 90^0

(c) 30^0 (d) 60^0

RRB JE - 23/05/2019 (Shift-III)

Ans : (a) $\frac{\cos\theta}{1-\sin\theta}-\frac{\cos\theta}{1+\sin\theta}=2$

$\Rightarrow\frac{\cos\theta(1+\sin\theta)-\cos\theta(1-\sin\theta)}{(1-\sin^2\theta)}=2$

$\Rightarrow\frac{\cos\theta+\cos\theta.\sin\theta-\cos\theta+\cos\theta.\sin\theta}{\cos^2\theta}=2$

$\Rightarrow\frac{2\cos\theta.\sin\theta}{\cos^2\theta}=2$ [∵ tan45° = 1]

$\Rightarrow$ tanθ = 1 = tan45°

$\Rightarrow$ θ = 45°

30. If x = r sin A cos B, y = r sin A sin B and z = r cos A, then find the value of $x^2+y^2+z^2$

(a) $r^2(\cos^2B+\cos^2A)$ (b) $2r^2$

(c) $3/2\ r^2$ (d) r^2

RRB JE - 24/05/2019 (Shift-II)

Ans : (d) x = r sin A cos B

y = r sin A sin B

z = r cos A

$x^2+y^2+z^2=(r\sin A\cos B)^2+(r\sin A\sin B)^2+(r\cos A)^2$

$=r^2\sin^2A\cos^2B+r^2\sin^2A\sin^2B+r^2\cos^2A$

$=r^2\sin^2A(\cos^2B+\sin^2B)+r^2\cos^2A$

$=r^2\sin^2A+r^2\cos^2A$ [$\sin^2B+\cos^2B=1$]

$=r^2(\sin^2A+\cos^2A)=r^2$

31. Simplify: sin (A+B) sin (A-B)

(a) $\sin^2A-\sin^2B$ (b) $\sin^2A+\sin^2B$

(c) cos 2A (d) $\cos^2A-\cos^2B$

RRB JE - 24/05/2019 (Shift-III)

Ans : (a) sin (A+B) sin (A–B)

formula (a + b) (a – b) = a^2-b^2

(sinA.cosB + cosA.sinB) (sinA.cosB – cosA.sinB)

$(\sin A.\cos B)^2-(\cos A.\sin B)^2$

$\sin^2A.\cos^2B-\sin^2B.\cos^2A$

$\sin^2A(1-\sin^2B)-\sin^2B(1-\sin^2A)$

$\sin^2A-\sin^2A.\sin^2B-\sin^2B+\sin^2A.\sin^2B$

$\boxed{\sin^2A-\sin^2B}$

32. Simplify: cos θ (1 – tan θ) + sin θ (1 – cot θ)

(a) sin θ + cos θ (b) sin θ – cos θ

(c) 0 (d) tan θ + cot θ

RRB JE - 27/05/2019 (Shift-I)

Ans : (c) cos θ (1 – tan θ) + sin θ (1 – cot θ)

$=\cos\theta\left(1-\frac{\sin\theta}{\cos\theta}\right)+\sin\theta\left(1-\frac{\cos\theta}{\sin\theta}\right)$

$=\cos\theta-\cos\theta\frac{\sin\theta}{\cos\theta}+\sin\theta-\sin\theta\frac{\cos\theta}{\sin\theta}$

= cosθ – sinθ + sinθ – cosθ = 0

33. If x = a sec θ + b tan θ and y = a tan θ + b sec θ, then find the value of x^2-y^2

(a) a^2+b^2 (b) $\sqrt{a^2+b^2}$

(c) a + b (d) a^2-b^2

RRB RPF SI – 12/01/2019 (Shift-II)

Ans : (d)

x = a secθ + b tanθ and y = a tanθ + b sec θ

$x^2-y^2=?$

$x^2-y^2=(a\sec\theta+b\tan\theta)^2-(a\tan\theta+b\sec\theta)^2$

$=a^2\sec^2\theta+b^2\tan^2\theta+2ab\sec\theta\tan\theta-a^2\tan^2\theta-b^2\sec^2\theta-2ab\sec\theta\tan\theta$

$=a^2\sec^2\theta+b^2\tan^2\theta-a^2\tan^2\theta-b^2\sec^2\theta$

$=a^2(\sec^2\theta-\tan^2\theta)-b^2(\sec^2\theta-\tan^2\theta)$

$x^2-y^2=(a^2-b^2)(\sec^2\theta-\tan^2\theta)$

$x^2-y^2=a^2-b^2$ $\{\sec^2\theta-\tan^2\theta=1$

34. Simplify:

$\frac{\sin\theta-2\sin^3\theta}{2\cos^3\theta-\cos\theta}$

(a) tan θ (b) sin θ – cos θ

(c) 2 sin θ cos θ (d) sin θ + cos θ

RRB JE - 31/05/2019 (Shift-I)

Ans : (a) $\frac{\sin\theta-2\sin^3\theta}{2\cos^3\theta-\cos\theta}=\frac{\sin\theta(1-2\sin^2\theta)}{\cos\theta(2\cos^2\theta-1)}$

$=\frac{\sin\theta.\cos2\theta}{\cos\theta.\cos2\theta}=\tan\theta$

35. Simplify:

$\frac{\tan A}{(1+\tan^2A)^2}+\frac{\cot A}{(1+\cot^2A)^2}$

(a) 2 sin A cos A (b) sin A-cos A

(c) sin A cos A (d) $(\sin A+\cos A)^2$

RRB JE - 31/05/2019 (Shift-III)

Ans. (c) $\frac{\tan A}{(1+\tan^2A)^2}+\frac{\cot A}{(1+\cot^2A)^2}$

$=\frac{\frac{\sin A}{\cos A}}{(\sec^2A)^2}+\frac{\frac{\cos A}{\sin A}}{(\text{cosec}^2A)^2}$

$=\frac{\sin A}{\cos A}\cos^4A+\frac{\cos A}{\sin A}\sin^4A$

$=\sin A.\cos^3A+\cos A.\sin^3A$

$=\sin A.\cos A(\cos^2A+\sin^2A)$

$=\sin A.\cos A$ (∵ $\sin^2A+\cos^2A=1$)

36. Simplify:
sin θ/(1 - cos θ)
(a) tan θ- sec θ (b) cosec θ + cot θ
(c) cosec θ -cot θ (d) tan θ + sec θ
RRB JE - 02/06/2019 (Shift-I)

Ans : (b) $\frac{\sin\theta}{1-\cos\theta}$

$= \frac{\sin\theta(1+\cos\theta)}{(1-\cos\theta)(1+\cos\theta)}$ (rationalization)

$= \frac{\sin\theta\,(1+\cos\theta)}{1-\cos^2\theta} = \frac{\sin\theta(1+\cos\theta)}{\sin^2\theta}$

$= \frac{1+\cos\theta}{\sin\theta} = \frac{1}{\sin\theta} + \frac{\cos\theta}{\sin\theta} = \text{cosec}\,\theta + \cot\theta$

37. Simplify:
$\frac{1}{(\sec\theta - \tan\theta)}$
(a) $\sec^2\theta - \tan^2\theta$ (b) $\cos\theta + \sin\theta$
(c) $\text{cosec}\,\theta - \cot\theta$ (d) $\sec\theta + \tan\theta$
RRB JE - 01/06/2019 (Shift-III)

Ans. (d)
$\frac{1}{(\sec\theta - \tan\theta)}$
Multiplying in denominator and numerator of (secθ + tanθ)

$\Rightarrow \frac{1(\sec\theta + \tan\theta)}{(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)}$

$\Rightarrow \frac{(\sec\theta + \tan\theta)}{(\sec^2\theta - \tan^2\theta)}$ $\{\sec^2\theta - \tan^2\theta = 1\}$

$\Rightarrow (\sec\theta + \tan\theta)$

38. Simplify: ($\sin^2\theta/\cos\theta$) +($\cos^2\theta/\cos\theta$)
(a) tan θ (b) cot θ
(c) cosec θ (d) sec θ
RRB JE - 28/06/2019 (Shift-III)

Ans. (d) $\frac{\sin^2\theta}{\cos\theta} + \frac{\cos^2\theta}{\cos\theta} = \left(\frac{\sin^2\theta + \cos^2\theta}{\cos\theta}\right)$

$\frac{1}{\cos\theta} = \sec\theta$

39. $\frac{\sin A + \sin B}{\cos A - \cos B} + \frac{\cos A + \cos B}{\sin A - \sin B} = ?$
(a) sin A cos B (b) 0
(c) tan A tan B (d) cos A cos B
RRB Group-D – 17/09/2018 (Shift-I)

Ans : (b)
$\frac{\sin A + \sin B}{\cos A - \cos B} + \frac{\cos A + \cos B}{\sin A - \sin B}$

$= \frac{(\sin A + \sin B)(\sin A - \sin B) + (\cos A + \cos B)(\cos A - \cos B)}{(\cos A - \cos B)(\sin A - \sin B)}$

$= \frac{\sin^2 A - \sin^2 B + \cos^2 A - \cos^2 B}{(\cos A - \cos B)(\sin A - \sin B)}$

$= \frac{(\sin^2 A + \cos^2 A) - (\sin^2 B + \cos^2 B)}{(\cos A - \cos B)(\sin A - \sin B)}$

$(\because \sin^2\theta + \cos^2\theta = 1)$

$= \frac{1-1}{(\cos A - \cos B)(\sin A - \sin B)} = 0$

40. Complete the following–
ΔABC, cos (B+C/2) = ?
(a) cos A (b) sin A/2
(c) sin A + B /2 (d) cot B
RRB Group-D – 29/10/2018 (Shift-III)

Ans : (b) $\because A + B + C = 180^0$
$B + C = 180 - A$

$\frac{B+C}{2} = \left(\frac{180 - A}{2}\right)$

$\cos\frac{(B+C)}{2} = \cos\frac{(180^0 - A)}{2}$

$\cos\left(\frac{B+C}{2}\right) = \cos\left(90^0 - \frac{A}{2}\right)$

$\cos\left(\frac{B+C}{2}\right) = \sin\frac{A}{2}$

41. If $2x = \sec A$ and $\left(\frac{2}{x}\right) = \tan A$, then find the value of $2\left(x^2 - \frac{1}{x^2}\right) =$
(a) 1 (b) 1/2
(c) 1/4 (d) 1/3
RRB Group-D – 10/10/2018 (Shift-III)

Ans : (b)
$2x = \sec A$ ..... (i)
$\frac{2}{x} = \tan A$ ..... (ii)
Squaring of equation (i) and (ii) then subtracting

$4x^2 - \frac{4}{x^2} = \sec^2 A - \tan^2 A$

$\{\sec^2 A - \tan^2 A = 1\}$

or $4\left(x^2 - \frac{1}{x^2}\right) = 1$

or $2\left(x^2 - \frac{1}{x^2}\right) = \frac{1}{2}$

42. In triangle ABC, find the value of $\tan^2\frac{A}{2} + \tan^2\frac{B}{2} + \tan^2\frac{C}{2}$
(a) > 1 (b) ≥ 1
(c) <1 (d) ≤ 1
RRB Group-D – 22/10/2018 (Shift-II)

Ans : (b) $\because$ in Δ ABC,
$A + B + C = \pi$
We know

$$\left(\tan\frac{A}{2}-\tan\frac{B}{2}\right)^2+\left(\tan\frac{B}{2}-\tan\frac{C}{2}\right)^2+\left(\tan\frac{C}{2}-\tan\frac{A}{2}\right)^2\geq 0$$

$$\Rightarrow 2\left(\tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2}\right)$$

$$-2\left(\tan\frac{A}{2}.\tan\frac{B}{2}+\tan\frac{B}{2}.\tan\frac{C}{2}+\tan\frac{C}{2}.\tan\frac{A}{2}\right)\geq 0....(I)$$

$A+B+C=\pi$

$$\frac{A+B+C}{2}=\frac{\pi}{2}$$

$$\frac{A+B}{2}=\frac{\pi}{2}-\frac{C}{2}$$

$$\tan\left(\frac{A}{2}+\frac{B}{2}\right)=\tan\left(\frac{\pi}{2}-\frac{C}{2}\right)$$

$$\frac{\tan\frac{A}{2}+\tan\frac{B}{2}}{1-\tan\frac{A}{2}\tan\frac{B}{2}}=\cot\frac{C}{2}$$

$$\frac{\tan\frac{A}{2}+\tan\frac{B}{2}}{1-\tan\frac{A}{2}\tan\frac{B}{2}}=\frac{1}{\tan\frac{C}{2}}$$

$$\tan\frac{A}{2}\tan\frac{C}{2}+\tan\frac{B}{2}\tan\frac{C}{2}=1-\tan\frac{A}{2}\tan\frac{B}{2}$$

$$\tan\frac{A}{2}\tan\frac{B}{2}+\tan\frac{B}{2}\tan\frac{C}{2}+\tan\frac{A}{2}\tan\frac{C}{2}=1.....(ii)$$

Putting value in equation (i) to equator (ii) -

$$2\left(\tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2}\right)-2\times 1\geq 0$$

$$\Rightarrow\left(\tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2}\right)\geq 1$$

43. If $x=r\cos\theta\cos\phi$, $y=r\cos\theta\sin\phi$ and $z=r\sin\theta$, then find the value of $x^2+y^2+z^2$

(a) y^2 (b) x^2
(c) r^2 (d) z^2

RRB Group-D – 26/10/2018 (Shift-II)

Ans : (c) given-

$x = r\cos\theta\cos\phi$ .....(1)
$y = r\cos\theta\sin\phi$ ..... (2)
$z = r\sin\theta$ .....(3)

Squaring of equation (1), (2) and (3) by Add-

$$x^2+y^2+z^2=r^2\cos^2\theta\cos^2\phi+r^2\cos^2\theta\sin^2\phi+r^2\sin^2\theta$$

$$=r^2\cos^2\theta\left[\cos^2\phi+\sin^2\phi\right]+r^2\sin^2\theta$$

$$=r^2\cos^2\theta+r^2\sin^2\theta \qquad \left[\because \sin^2\phi+\cos^2\phi=1\right]$$

$$=r^2\left(\sin^2\theta+\cos^2\theta\right)$$

$$\boxed{x^2+y^2+z^2=r^2}$$

44. Find the value of $(\sec\theta-\tan\theta)^2$

(a) $\cot\theta$
(b) $(1-\sin\theta)/(1+\sin\theta)$
(c) $\cos\theta\times\operatorname{cosec}\theta$
(d) 1

RRB NTPC 04.04.2016 Shift : 2

Ans : (b) $(\sec\theta-\tan\theta)^2=\left(\frac{1}{\cos\theta}-\frac{\sin\theta}{\cos\theta}\right)^2$

$$=\left(\frac{1-\sin\theta}{\cos\theta}\right)^2=\frac{(1-\sin\theta)^2}{\cos^2\theta}$$

$$=\frac{(1-\sin\theta)^2}{1-\sin^2\theta} \qquad (\because \sin^2\theta+\cos^2\theta=1)$$

$$=\frac{(1-\sin\theta)(1-\sin\theta)}{(1-\sin\theta)(1+\sin\theta)}=\frac{1-\sin\theta}{1+\sin\theta}$$

45. $\sin^6A + \cos^6A - 1$ is equal to

(a) $-3\sin^2A\cos^2A$ (b) $1-3\sin A\cos A$
(c) $1+3\sin^2A\cos^2A$ (d) 0

RRB NTPC 18.04.2016 Shift : 1

Ans : (a)

$\sin^6A + \cos^6A-1$
$= (\sin^2A)^3 + (\cos^2A)^3-1$

Formula,

Form, $a^3+b^3 = (a+b)^3-3ab\,(a+b)$

$= (\sin^2A+\cos^2A)^3-3\sin^2A.\cos^2A(\sin^2A+\cos^2A)-1$

$= (1)^3-3\sin^2A.\cos^2A(1)-1 \quad \{\sin^2 A+\cos^2 A=1\}$

$= 1-3\sin^2A.\cos^2A-1$

$= -3\sin^2A.\cos^2A$

46. Simplify:$(\sin\theta / \cos\theta) \times (\cot\theta / \operatorname{cosec}\theta)$

(a) $\cos\theta$ (b) $\sin\theta$
(c) $\tan\theta$ (d) $\sec\theta$

RRB NTPC 06.04.2016 Shift : 2

Ans : (b) $\frac{\sin\theta}{\cos\theta}\times\frac{\cot}{\operatorname{cosec}\theta}$

$$\frac{\sin\theta}{\cos\theta}\times\frac{\frac{\cos\theta}{\sin\theta}}{\frac{1}{\sin\theta}}=\sin\theta$$

47. What is the value of $\sin\theta\tan\theta - \sec\theta$

(a) $-\cos\theta$ (b) 1
(c) $-\sec\theta$ (d) $\operatorname{cosec}\theta$

RRB NTPC 28.04.2016 Shift : 1

Ans : (a) $\sin\theta\tan\theta - \sec\theta=\sin\theta\frac{\sin\theta}{\cos\theta}-\frac{1}{\cos\theta}$

$$=\frac{\sin^2\theta}{\cos\theta}-\frac{1}{\cos\theta}$$

$$=\frac{\sin^2\theta}{\cos\theta}-\left(\frac{\sin^2\theta+\cos^2\theta}{\cos\theta}\right) \qquad \{1=\sin^2\theta+\cos^2\theta\}$$

$$=\frac{\sin^2\theta-\sin^2\theta-\cos^2\theta}{\cos\theta}=\frac{-\cos^2\theta}{\cos\theta}=-\cos\theta$$

48. Solve the following :

$$\frac{1}{1+\sin\theta}+\frac{1}{1-\sin\theta}=?$$

(a) 0 (b) $2\,\text{Cos}^2\theta$
(c) $2\text{Sec}^2\theta$ (d) 1

RRB ALP & Tec. (17-08-18 Shift-III)

Ans : (c) $\frac{1}{1+\sin\theta}+\frac{1}{1-\sin\theta}=\frac{1-\sin\theta+1+\sin\theta}{1-\sin^2\theta}$

$=\frac{2}{1-\sin^2\theta}$ $\left[\because 1-\sin^2\theta=\cos^2\theta\right]$

$=\frac{2}{\cos^2\theta}=2\sec^2\theta$

Type - 3

49. If sin (A – B) = sin A cos B – cos A sin B, then what is the value of sin15°?

(a) $\frac{\sqrt{3}-1}{4}$ (b) $\frac{\sqrt{6}-\sqrt{2}}{2}$
(c) $\frac{\sqrt{3}-\sqrt{2}}{2}$ (d) $\frac{\sqrt{6}-\sqrt{2}}{4}$

RRB RPF Constable – 17/01/2019 (Shift-I)

Ans : (d) sin (A – B) = sinA cosB – cosA sinB

$\sin 15^0=\sin(60°-45°)=\sin 60^0\cos 45^0-\cos 60^0\sin 45^0$

$\sin 15^0=\frac{\sqrt{3}}{2}\times\frac{1}{\sqrt{2}}-\frac{1}{2}\times\frac{1}{\sqrt{2}}$

$=\frac{\sqrt{3}}{2\sqrt{2}}-\frac{1}{2\sqrt{2}}$

$\sin 15^0=\frac{\sqrt{3}-1}{2\sqrt{2}}$

$\sin 15^0=\frac{(\sqrt{3}-1)\sqrt{2}}{2\times 2}=\frac{\sqrt{6}-\sqrt{2}}{4}$

50. The value of Cos(-780^0) is

(a) $\sqrt{3}/2$ (b) $-\sqrt{3}/2$
(c) $1/2$ (d) $-1/2$

RRB RPF SI – 13/01/2019 (Shift-II)

Ans : (c) Cos(-780^0)

= Cos 780^0 [$\because$ Cos (-θ) = Cosθ]

= Cos (2 × 360^0 + 60^0)

= Cos 60^0 = $\frac{1}{2}$

51. If tan 2A = cot(A-18^0) and 2A is acute angle, then find the value of 'A'

(a) 36^0 (b) 24^0
(c) 28^0 (d) 18^0

RRB JE - 22/05/2019 (Shift-III)

Ans : (a) tan2A = cot (A – 18°)

cot (90° – 2A) = cot (A – 18°)

90° – 2A = A – 18° [$\because$ cot (90° – θ) = tanθ]

108° = 3A

A = 36°

52. Find the value of sin 75°?

(a) $\frac{\sqrt{6}+\sqrt{2}}{4}$ (b) $\frac{\sqrt{6}-\sqrt{2}}{4}$
(c) $\frac{\sqrt{3}-1}{2\sqrt{2}}$ (d) $\frac{\sqrt{3}+1}{2}$

RRB JE - 23/05/2019 (Shift-III)

Ans : (a) sin (75°)

= sin (45° + 30°)

= sin45°. cos30° + cos45°. sin30°

$=\frac{1}{\sqrt{2}}\times\frac{\sqrt{3}}{2}+\frac{1}{\sqrt{2}}\times\frac{1}{2}$

$=\frac{\sqrt{3}}{2\sqrt{2}}+\frac{1}{2\sqrt{2}}=\frac{\sqrt{3}+1}{2\sqrt{2}}\times\frac{\sqrt{2}}{\sqrt{2}}$

$=\frac{\sqrt{6}+\sqrt{2}}{4}$

53. tan (60° + 30°) – tan (60° – 30°) = ?

(a) $\frac{1}{2}$ (b) $\frac{3}{2}$
(c) Infinite (d) 1

RRB Group-D – 30/10/2018 (Shift-I)

Ans : (c) tan (60 + 30) – tan (60 – 30)

= tan 90° - tan 30°

$=\frac{1}{0}-\frac{1}{\sqrt{3}}=\frac{\sqrt{3}-0}{0}=\frac{\sqrt{3}}{0}=\infty$

54. $\frac{2\sin 30°}{1+\cos 30°}+\frac{1+\cos 30°}{\sin 30°}=?$

(a) 4 (b) 8
(c) $4-2\sqrt{3}$ (d) $6-\sqrt{3}$

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (d) $\frac{2\sin 30°}{1+\cos 30°}+\frac{1+\cos 30°}{\sin 30°}$

$=\frac{2\sin^2 30°+(1+\cos 30°)^2}{\sin 30°(1+\cos 30°)}=\frac{2\times\frac{1}{4}+\left(1+\frac{\sqrt{3}}{2}\right)^2}{\frac{1}{2}\left(1+\frac{\sqrt{3}}{2}\right)}$

$=\frac{\frac{1}{2}+\frac{(2+\sqrt{3})^2}{4}}{\frac{1}{2}\times\frac{(2+\sqrt{3})}{2}}=\frac{2+7+4\sqrt{3}}{2+\sqrt{3}}=\frac{9+4\sqrt{3}}{2+\sqrt{3}}$

$=\frac{9+4\sqrt{3}}{2+\sqrt{3}}\times\frac{2-\sqrt{3}}{2-\sqrt{3}}$

$=\frac{18+8\sqrt{3}-9\sqrt{3}-12}{4-3}=6-\sqrt{3}$

55. $\frac{\tan 45^0}{1+\cos 45^0}+\frac{1+\sin 45^0}{\cot 45^0}=?$

(a) $3-\frac{\sqrt{2}}{2}$ (b) $\sqrt{2}$

(c) $3-2\sqrt{2}$ (d) $-\frac{\sqrt{2}}{2}$

RRB Group-D – 30/10/2018 (Shift-III)

Ans. (a) :

$$\frac{\tan 45^0}{1+\cos 45^0}+\frac{1+\sin 45^0}{\cot 45^0}$$

$$\frac{1}{1+\frac{1}{\sqrt{2}}}+\frac{1+\frac{1}{\sqrt{2}}}{1}$$

$$\frac{\sqrt{2}}{\sqrt{2}+1}+\frac{\sqrt{2}+1}{\sqrt{2}}=\frac{2+(\sqrt{2}+1)^2}{\sqrt{2}(\sqrt{2}+1)}=\frac{2+2+1+2\sqrt{2}}{\sqrt{2}(\sqrt{2}+1)}$$

$$\frac{5+2\sqrt{2}}{\sqrt{2}(\sqrt{2}+1)}$$

$$\frac{\left(5+2\sqrt{2}\right)}{\sqrt{2}\left(\sqrt{2}+1\right)}\times\frac{\sqrt{2}\left(\sqrt{2}-1\right)}{\sqrt{2}\left(\sqrt{2}-1\right)}=\frac{10+4\sqrt{2}-5\sqrt{2}-4}{2(2-1)}$$

$$=\frac{6-\sqrt{2}}{2}=3-\frac{\sqrt{2}}{2}$$

56. Which of the following is $\sin(90°-\theta)$

(a) cosec θ (b) cot θ

(c) sin θ (d) cos θ

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (d) : $\sin(90-\theta)=\cos\theta$

57. Given that $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$, then what is the value of $\tan 75^0$?

(a) $2+2\sqrt{3}$ (b) $2-\sqrt{3}$

(c) $2+\sqrt{3}$ (d) $2-2\sqrt{3}$

RRB Group-D – 27/11/2018 (Shift-I)

Ans. (c) : on putting A = 45° and B = 30°,

$$\Rightarrow\frac{\tan 45°+\tan 30°}{1-\tan 45°\tan 30°}=\tan(45°+30°)$$

$$\Rightarrow\frac{\tan 45°+\tan 30°}{1-\tan 45°\tan 30°}=\tan 75°$$

$$\Rightarrow\frac{1+\frac{1}{\sqrt{3}}}{1-1\times\frac{1}{\sqrt{3}}}=\tan 75°$$

$$\Rightarrow\frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}}=\frac{\sqrt{3}+1}{\sqrt{3}-1}=\tan 75°$$

rationalization,

$$\Rightarrow\frac{\sqrt{3}+1}{\sqrt{3}-1}\times\frac{\sqrt{3}+1}{\sqrt{3}+1}$$

$$\Rightarrow\frac{\left(\sqrt{3}+1\right)^2}{\left(\sqrt{3}-1\right)\left(\sqrt{3}+1\right)}$$

$$\Rightarrow\frac{\left(\sqrt{3}+1\right)^2}{\left(\sqrt{3}-1\right)\left(\sqrt{3}+1\right)}=\frac{\left(\sqrt{3}\right)^2+(1)^2+2\times\sqrt{3}\times 1}{\left(\sqrt{3}\right)^2-(1)^2}$$

formula–

$\because\ (a+b)^2=a^2+b^2+2ab$ and

$a^2-b^2=(a+b)(a-b)$ from,

$$=\frac{3+1+2\sqrt{3}}{3-1}=\frac{4+2\sqrt{3}}{2}=\tan 75°$$

$$=\frac{2\left(2+\sqrt{3}\right)}{2}=\tan 75°=2+\sqrt{3}=\tan 75°$$

58. If sin15° = $\frac{\sqrt{6}-\sqrt{2}}{4}$, then approx value of tan15° ?

(a) $2-2\sqrt{3}$ (b) $2+2\sqrt{3}$

(c) $2-\sqrt{3}$ (d) $2+2\sqrt{3}$

Ans. (c) : sin 15° = $\frac{\sqrt{6}-\sqrt{2}}{4}$

following value sin15° = sin (45° – 30°) got the solution by formula sin (A – B). Likewise for tan 15° = tan (45° – 30°) will used of tan (A – B)

$$\tan(45°-30°)=\frac{\tan 45°-\tan 30°}{1+\tan 45°\tan 30°}$$

$$=\frac{1-\frac{1}{\sqrt{3}}}{1+1\times\frac{1}{\sqrt{3}}}=\frac{\frac{\sqrt{3}-1}{\sqrt{3}}}{\frac{\sqrt{3}+1}{\sqrt{3}}}=\frac{\sqrt{3}-1}{\sqrt{3}+1}\times\frac{\sqrt{3}-1}{\sqrt{3}-1}$$

$$=\frac{\left(\sqrt{3}-1\right)^2}{\left(\sqrt{3}\right)^2-(1)^2}=\frac{3+1-2\sqrt{3}}{3-1}$$

$$=\frac{4-2\sqrt{3}}{2}=\frac{2\left(2-\sqrt{3}\right)}{2}=2-\sqrt{3}$$

59. If $\text{coses}(A-B)=\frac{\sec A\sec B}{\tan A-\tan B}$, then coses 15°= ?

(a) $\sqrt{6}-\sqrt{3}$ (b) $\sqrt{6}-\sqrt{2}$

(c) $\sqrt{6}+\sqrt{3}$ (d) $\sqrt{6}+\sqrt{2}$

RRB Group-D – 05/11/2018 (Shift-III)

Ans. (d) :

Suppose

A = 60°

B = 45°

$$\text{cosec}(60°-45°)=\frac{\sec 60°\times\sec 45°}{\tan 60°-\tan 45°}$$

$$\operatorname{cosec}15° = \frac{2\times\sqrt{2}}{\sqrt{3}-1}$$

$$= \frac{2\sqrt{2}}{\sqrt{3}-1}\times\frac{(\sqrt{3}+1)}{(\sqrt{3}+1)} = \frac{2\sqrt{2}(\sqrt{3}+1)}{(\sqrt{3})^2-(1)^2}$$

$$= \frac{2\sqrt{6}+2\sqrt{2}}{3-1} = \frac{2\sqrt{6}+2\sqrt{2}}{2} = \frac{2(\sqrt{6}+\sqrt{2})}{2}$$

$$\operatorname{cosec}15° = \sqrt{6}+\sqrt{2}$$

60. Identify the in compatible:

$\sin 90^0$	$\cos 0^0$	$\tan 90^0$	$\tan 45^0$
A	B	C	D

(a) B (b) C
(c) A (d) D

RRB Group-D – 23/10/2018 (Shift-III)

Ans : (b) Sin 90° = 1
Cos0° = 1
tan 90°= ∞
and tan 45° = 1
tan 90°
tan 90° is inconsistent because trigonometry ratio got the value 1 in other when tan 90° is undefined (∞)

61. Find the value of

$$\frac{5\cos^2 60^0 + 4\sec^2 30^0 - \tan^2 45^0}{\sin^2 30^0 + \cos^2 30^0} = ?$$

(a) 1 (b) 2
(c) $\frac{49}{12}$ (d) $\frac{67}{12}$

RRB Group-D – 09/10/2018 (Shift-II)

Ans. (d) : $\frac{5\cos^2 60^0 + 4\sec^2 30^0 - \tan^2 45^0}{\sin^2 30^0 + \cos^2 30^0}$

$\because \sin^2\theta + \cos^2\theta = 1$

$$= \frac{5\times\left(\frac{1}{2}\right)^2 + 4\times\left(\frac{2}{\sqrt{3}}\right)^2 - (1)^2}{1} = \frac{5\times\frac{1}{4}+4\times\frac{4}{3}-1}{1}$$

$$= \frac{5}{4}+\frac{16}{3}-1 = \frac{15+64-12}{12} = \frac{67}{12}$$

62. $\frac{(\cos 60^0 + \sin 60^0)}{(\cos 60^0 - \sin 60^0)} = ?$

(a) $-(\sqrt{3}+2)$ (b) $(\sqrt{3}-2)$
(c) -1 (d) $-\frac{\sqrt{3}}{2}$

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (a) $\left(\frac{\cos 60° + \sin 60°}{\cos 60° - \sin 60°}\right)$

$$= \frac{\left(\frac{1}{2}+\frac{\sqrt{3}}{2}\right)}{\left(\frac{1}{2}-\frac{\sqrt{3}}{2}\right)} = \frac{\frac{1+\sqrt{3}}{2}}{\frac{1-\sqrt{3}}{2}} = \frac{1+\sqrt{3}}{1-\sqrt{3}}$$

$$= \frac{(1+\sqrt{3})(1+\sqrt{3})}{(1-\sqrt{3})(1+\sqrt{3})}$$

Multiplying in denominator and numerator from $(1+\sqrt{3})$

$$= \frac{(1+\sqrt{3})^2}{(1)^2-(\sqrt{3})^2} \Rightarrow \frac{1+3+2\sqrt{3}}{1-3}$$

$$= \frac{1+3+2\sqrt{3}}{-2}, \quad = \frac{(4+2\sqrt{3})}{-2}$$

$$= \frac{2(2+\sqrt{3})}{-2}, \quad = -(\sqrt{3}+2)$$

63. Find the value of $[\sin(45^0) + \cos(45^0)]^2$

(a) 1/2 (b) $1/\sqrt{2}$
(c) 3/2 (d) 2

RRB NTPC 05.04.2016 Shift : 2

Ans : (d) $[\sin(45^0) + \cos(45^0)]^2$

$$= \left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right)^2$$

$$= \left(\frac{2}{\sqrt{2}}\right)^2 = \frac{4}{2} = 2$$

64. If cot 52° = b, tan 38° = ?

(a) $\sqrt{b}$ (b) $\sqrt{b}/2$
(c) -b (d) b

RRB NTPC 05.04.2016 Shift : 3

Ans :(d) $\cot 52^0 = b$

$\cot(90^0 - 38^0) = b$

$\tan 38^0 = b \quad [\because \cot(90^0-\theta) = \tan\theta]$

65. What is the value of tan (315°)?

(a) 1 (b) -1
(c) $1/\sqrt{2}$ (d) $-1/\sqrt{2}$

RRB NTPC 04.04.2016 Shift : 3

Ans : (b) $\tan 315^0 = \tan(360^0 - 45^0)$

$= -\tan 45^0 \quad [\because \tan(360^0-\theta) = -\tan\theta]$

$= -1$

66. Find the value of $\tan(1125^0)$ is

(a) 1 (b) −1
(c) 0 (d) ∞

RRB NTPC 31.03.2016 Shift : 1

Ans : (a) $\tan(1125^0)$
$= \tan(3\times 360^0 + 45^0)$
$= \tan 45^0 \quad [\because \tan(n\times 360^0+\theta) = \tan\theta]$
$= 1$

67. Find the value of tan (-405^0)

(a) 1 (b) –1
(c) ∞ (d) 0

RRB NTPC 29.03.2016 Shift : 3

Ans : (b) $\tan(-405^0)$
$= -\tan 405^0$
$= -\tan(360^0 + 45^0)$
$= -\tan 45^0 = -1$
$[\because \tan(360^0+\theta) = \tan\theta]$

68. $\text{cosec}(90^0 - \theta) = ?$

(a) $\tan\theta$ (b) $\cot\theta$
(c) $\sec\theta$ (d) $\cos\theta$

RRB NTPC 06.04.2016 Shift : 1

Ans : (c) $\text{cosec}(90-\theta) = \sec\theta$

69. Find the value of cos0° is:

(a) 0 (b) Infinite
(c) Not defined (d) 1

RRB NTPC 27.04.2016 Shift : 3

Ans : (d) $\cos 0^0 = 1$,

70. Evaluate: $\frac{\sin 23^0}{\cos 67^0}$

(a) 1/2 (b) 2
(c) 1 (d) 2/3

RRB NTPC 29.04.2016 Shift : 1

Ans : (c) $\frac{\sin 23^0}{\cos 67^0} = \frac{\sin 23^0}{\cos(90^\circ - 23)} = \frac{\sin 23^\circ}{\sin 23^\circ} = 1$

71. Find the value of $\text{cosec}\, 60^0$

(a) $\sqrt{3}/2$ (b) $\sqrt{2}$
(c) $2/\sqrt{3}$ (d) $1/\sqrt{2}$

RRB NTPC 30.04.2016 Shift : 1

Ans : (c) $\text{cosec}\, 60^0 = \frac{1}{\sin 60^0} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$

Type - 4

72. Find the value of $\sin 120^0 \sin 240^0 \sin 270^0$

(a) -1/8 (b) -1/2
(c) 3/4 (d) 1/8

RRB JE - 22/05/2019 (Shift-I)

Ans : (c) sin120° sin240° sin270°
⇒ sin (90° + 30°) sin (180° + 60°) sin (180° + 90°)
⇒ cos30° (–sin60°) (–sin90°)
$\Rightarrow \frac{\sqrt{3}}{2} \times \left(-\frac{\sqrt{3}}{2}\right) \times (-1) = \frac{3}{4}$

73. Find the value of $\sin\frac{7\pi}{4}\sin\frac{\pi}{4}\sin\frac{3\pi}{4}\sin\frac{5\pi}{4}$

(a) $\frac{1}{4}$ (b) $\frac{1}{8}$
(c) $\frac{1}{16}$ (d) $\frac{3}{16}$

RRB JE - 22/05/2019 (Shift-I)

Ans : (a)

$$\sin\frac{7\pi}{4}\sin\frac{\pi}{4}\sin\frac{3\pi}{4}\sin\frac{5\pi}{4}$$
$$= \sin\left(\pi + \frac{3\pi}{4}\right)\sin\frac{\pi}{4}\sin\frac{3\pi}{4}\sin\left(\pi + \frac{\pi}{4}\right)$$
$$= \left(-\sin\frac{3\pi}{4}\right)\sin\frac{\pi}{4}\sin\frac{3\pi}{4}\left(-\sin\frac{\pi}{4}\right)$$
$$= \sin\frac{3\pi}{4}\sin\frac{\pi}{4}\sin\frac{3\pi}{4}\sin\frac{\pi}{4}$$
$$= \sin\left(\pi - \frac{\pi}{4}\right)\sin\frac{\pi}{4}\sin\left(\pi - \frac{\pi}{4}\right)\sin\frac{\pi}{4}$$
$$= \sin\frac{\pi}{4}\sin\frac{\pi}{4}\sin\frac{\pi}{4}\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{4}$$

74. Simplify: cos18° + cos 162°+sin 126°+ sin 234°

(a) 2 (b) 1
(c) -2 (d) 0

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans : (d) $\cos 18^0 + \cos 162^0 + \sin 126^\circ + \sin 234^0$
= cos18°+cos (180°–18°) + sin 126° + sin (360° – 126°)
= cos 18° + (–cos 18°) + (sin 126°) + (–sin 126°)
= cos 18° – cos 18°+ sin 126° – sin 126° = 0

75. If $\tan^2 45^0 - \cos^2 60^0 = x \sin 45^0 \cos 45^0 \cot 30^0$, Find the value of 'x'

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{\sqrt{2}}$
(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{3}{2}$

RRB JE - 24/05/2019 (Shift-I)

Ans : (a) $\tan^2 45^0 - \cos^2 60^0 = x \sin 45^0 \cos 45^0 \cot 30^0$

$$1 - \left(\frac{1}{2}\right)^2 = x \times \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \cdot \sqrt{3}$$
$$\frac{3}{4} = \frac{x}{2} \times \sqrt{3}$$
$$x = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3} \times \sqrt{3}}{2 \times \sqrt{3}} = \frac{\sqrt{3}}{2}$$

76. Simplify: $\sin 780^0 \sin 480^0 + \cos 120^0 \sin 30^0$

(a) 2/3 (b) 1/3
(c) 0 (d) 1/2

RRB JE - 24/05/2019 (Shift-I)

Ans : (d) $\sin 780^0 \sin 480^0 + \cos 120^0 \sin 30^0$
= sin (2×360°+60°) sin (360° + 120°) + cos120° sin30°
= sin60° sin (120°) + cos120° sin30°
= sin60° sin(90° + 30°) + cos (90° + 30°) sin30°
= sin60° cos30° – sin30°sin30°
$= \frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \times \frac{1}{2} = \frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$

77. Simplify:
$\cos 5^0 + \cos 24^0 + \cos 175^0 + \cos 204^0 + \cos 300^0$
(a) 1/2 (b) 1
(c) -1/2 (d) 0

RRB JE - 25/05/2019 (Shift-I)

Ans : (a)
$\cos 5^0 + \cos 24^0 + \cos 175^0 + \cos 204^0 + \cos 300^0$
$= \cos 5° + \cos 24° + \cos (180° - 5°) + \cos (180° + 24°) + \cos (270° + 30°)$
$= \cos 5° + \cos 24° - \cos 5° - \cos 24° + \sin 30°$
$= \sin 30° = \frac{1}{2}$

78. Find the value of tan 10° tan 15° tan 80° tan 75°
(a) 1/3 (b) 1
(c) 1/2 (d) 2/3

RRB JE - 25/05/2019 (Shift-II)

Ans : (b) tan10°. tan 15° . tan 80°. tan75°
= tan10°. tan 15° . tan(90°–10°) tan (90°–15°)
= tan10°. tan 15°. cot 10°. cot 15°
$= \tan 10°.\tan 15° \times \frac{1}{\tan 10°} \times \frac{1}{\tan 15°}$
= 1

79. Find the value of $\tan^2 60^o - 2\tan^2 45^o - \cot^2 30^o + 2\sin^2 30^o + 3/4\ \text{cosec}^2 45^o$
(a) $-\frac{\sqrt{3}}{2}$ (b) 0
(c) 2 (d) – 1

RRB JE - 31/05/2019 (Shift-I)

Ans : (b) $\tan^2 60^o - 2\tan^2 45^o - \cot^2 30^o + 2\sin^2 30^o + 3/4\ \text{cosec}^2 45^o$
$(\sqrt{3})^2 - 2 \times 1 - (\sqrt{3})^2 + 2\left(\frac{1}{2}\right)^2 + \frac{3}{4} \times \left(\sqrt{2}\right)^2$
$3 - 2 - 3 + 2 \times \frac{1}{4} + \frac{3}{4} \times 2$
$3 - 5 + \frac{1}{2} + \frac{3}{2}$
$\frac{6 - 10 + 1 + 3}{2} = \frac{10 - 10}{2} = 0$

80. tan 34° tan42° tan48° tan56° + tan60° cot 30° - cosec30° sec60° = ?
(a) 0 (b) 1.5
(c) 1 (d) 2

RRB Group-D – 26/09/2018 (Shift-III)

Ans : (a) tan 34° tan42° tan48° tan56° + tan60° cot 30° - cosec30° sec60°
= tan (90°-56°) tan(90°-48°) tan48° tan56° + tan60° cot (90°-60°) - cosec(90°-60°) sec60°
= Cot 56° cot 48° tan 48° tan 56° + $\tan^2 60°$ - $\sec^2 60°$
$= 1 \times 1 + (-1) = 1 - 1 = 0$ $\left\{\begin{matrix} \tan\theta.\cot\theta = 1 \\ \sec^2\theta - \tan^2\theta = 1 \end{matrix}\right\}$

81. Find the value of $\tan 8^0\ \tan 22^0\ \cot 60^0\ \tan 68^0\ \tan 82^0$
(a) 1 (b) $\frac{2}{\sqrt{3}}$
(c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{3}$

RRB Group-D – 16/10/2018 (Shift-II)

Ans : (c)
$\tan 8^0 \tan 22^0 \cot 60^0 \tan 68^0 \tan 82^0$
$= \tan 8^0.\tan 22^0 \cot 60^0 \tan\left(90^0 - 22^0\right).\tan\left(90^0 - 8^0\right)$
$= \tan 8^0.\tan 22^0 \cot 60^0.\cot 22^0.\cot 8^0$
$= \tan 8^0 \times \frac{1}{\tan 8^0} \times \tan 22^0 \times \frac{1}{\tan 22^0}.\cot 60^0$
$= 1 \times 1 \times \cot 60^0$
$= 1 \times 1 \times \frac{1}{\sqrt{3}}$ $\left[\because \cot 60^0 = \frac{1}{\sqrt{3}}\right]$
$= \frac{1}{\sqrt{3}}$

82. Find the value of $\sin 12^0 \times \cos 18^0 \times \sec 78^0 \times \text{cosec}\ 72^0$
(a) $\frac{1}{2}$ (b) $\sqrt{3}$
(c) $\frac{\sqrt{3}}{3}$ (d) 1

Ans. (d) : $\sin 12^0 \times \cos 18^0 \times \sec 78^0 \times \text{cosec}\ 72^0$
$\sin 12^0 \times \cos 18^0 \times \frac{1}{\cos 78^0} \times \frac{1}{\sin 72^0}$
$= \frac{\sin 12^0 \times \cos 18^0}{\cos 78^0 \times \sin 72^0}$
$= \frac{\sin(90 - 78)^0 \times \cos(90 - 72)^0}{\cos 78^0 \times \sin 72^0}$
$= \frac{\cos 78^0 \times \sin 72^0}{\cos 78^0 \times \sin 72^0} = 1$

83. What is the sum of the value of all six trigonometric ratio when the angle $\alpha = 45^0$?
(a) $2 + 3\sqrt{2}$ (b) $\frac{6 + 3\sqrt{2}}{2}$
(c) 6 (d) $2 + 4\sqrt{2}$

RRB Group-D – 23/10/2018 (Shift-I)

Ans. (a) : Sum of all six trigonometry ratio
$= \sin\alpha + \cos\alpha + \tan\alpha + \cot\alpha + \sec\alpha + \text{cosec}\alpha$
$= \sin 45^0 + \cos 45^0 + \tan 45^0 + \cot 45^0 + \text{Sec}45^0 + \text{cosec}45^0$
$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + 1 + 1 + \sqrt{2} + \sqrt{2} = \frac{2}{\sqrt{2}} + 2 + 2\sqrt{2}$
$= \sqrt{2} + 2 + 2\sqrt{2} = 3\sqrt{2} + 2 = 2 + 3\sqrt{2}$

84. $\sin^2 60^0 + \cos^2 30^0 + \cot^2 45^0 + \sec^2 60^0 = ?$
(a) $\frac{7}{2}$ (b) $\frac{5}{2}$
(c) $\frac{13}{2}$ (d) $\frac{15}{2}$

RRB Group-D – 18/09/2018 (Shift-I)

Ans. (c) : $\sin^2 60^0 + \cos^2 30^0 + \cot^2 45^0 + \sec^2 60^0 = ?$

$$= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + (1)^2 + (2)^2$$

$$= \frac{3}{4} + \frac{3}{4} + 1 + 4 = \frac{6}{4} + 5 = \frac{6+20}{4} = \frac{26}{4} = \frac{13}{2}$$

85. What is the value of following expression? ($Cot1^0\ Cot2^0\ Cot3^0\ Cot4^0\ Cot5^0$ ….. $Cot90^0$)

(a) 0 (b) 1
(c) 2 (d) 1/2

RRB NTPC 28.04.2016 Shift : 3

Ans : (a) ($Cot1^0\ Cot2^0\ Cot3^0\ Cot4^0\ Cot5^0$ ….. $Cot90^0$)
$\because Cot\ 90^0 = 0$
So ($Cot1^0\ Cot2^0\ Cot3^0\ Cot4^0\ Cot5^0$ .. $Cot89^0 \times 0$) = 0

86. The value of $\cos 100^0 \cos 10^0 + \sin 100^0 \sin 10^0$ is equal to?

(a) 0 (b) $\cos 110^0$
(c) $\sin 110^0$ (d) 1

RRB NTPC 10.04.2016 Shift : 3

Ans : (a) cos (A–B) = cosA. cos B + sin A. sin B
from the question
$\cos 100^0 . \cos 10^0 + \sin 100^0 . \sin 10^0$
$= \cos (100^0 - 10^0) = \cos 90^0 = 0$

87. Find the value of tan24. tan48. tan42.tan66

(a) 0 (b) 1
(c) 1/2 (d) 2

RRB NTPC 22.04.2016 Shift : 1

Ans : (b) $\tan 24^0 . \tan 48^0 . \tan 42^0 . \tan 66^0$
$= \tan 24^0 . \tan 48^0 . \tan (90^0 - 48^0) . \tan (90^0 - 24^0)$
$= \tan 24^0 . \tan 48^0 . \cot 48^0 . \cot 24^0$
$\{\because \tan(90^0 - \theta) = \cot\theta\}$
= 1

88. What is the value of the following expression ($\tan 0^o\ \tan 1^o\ \tan 2^o\ \tan 3^o\ \tan 4^o$……$\tan 89^o$)

(a) 0 (b) 1
(c) 2 (d) 1/2

RRB NTPC 28.04.2016 Shift : 2

Ans : (a) $\because \tan 0^o = 0$
$\therefore \tan 0^o\ \tan 1^o\ \tan 2^o\ \tan 3^o$ ……$\tan 89^o = 0$

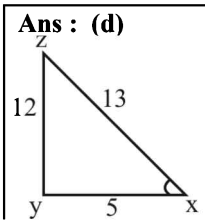
Type - 5

89. If $Cot\ x = \frac{5}{12}$, then sin x – sec x = ?

(a) $-\frac{229}{65}$ (b) $\frac{229}{65}$
(c) $\frac{109}{65}$ (d) $-\frac{109}{65}$

RRB RPF SI – 05/01/2019 (Shift-II)

Ans : (d)



$\because Cot\ x = \frac{xy}{yz} = \frac{5}{12}$

In right angle Δ xyz from Pythagoras theorem
$(hypotenuse)^2 = (perpendicular)^2 + (base)^2$
$(xz)^2 = (yz)^2 + (xy)^2$
$(xz)^2 = (12)^2 + (5)^2$
$(xz)^2 = 144 + 25$
$xz = \sqrt{169}$
xz = 13
∴ Sin x –Sec x
$= \frac{12}{13} - \frac{13}{5} = \frac{60-169}{65} = -\frac{109}{65}$

90. If cos A = 12/13, then sin A(1-tanA) = ?

(a) 32/135 (b) 32/137
(c) 35/152 (d) 35/156

RRB RPF SI – 11/01/2019 (Shift-III)

Ans : (d) Given-

$\cos A = \frac{12}{13} = \frac{base}{hypotenuse}$

$\therefore (perpendicular)^2 = (hypotenuse)^2 - (base)^2$
$= (13)^2 - (12)^2 = 25$

So perpendicular $= \sqrt{25} = 5$

$\therefore \sin A.(1-\tan A) = \frac{5}{13}\left(1 - \frac{5}{12}\right)$ $\{\because \sin\theta = \frac{Perpendicular}{hypotenuse}$

$\tan\theta = \frac{Perpendicular}{base}\}$

$= \frac{5}{13} \times \frac{7}{12} = \frac{35}{156}$

91. If tan α =1/2, tan $\beta = \frac{1}{3}$ then find the value of α + β

(a) 0^0 (b) 135^0
(c) 90^0 (d) 45^0

RRB JE - 24/05/2019 (Shift-III)

Ans : (d) $\tan\alpha = \frac{1}{2}$ $\tan\beta = \frac{1}{3}$ $\alpha + \beta = ?$

$$\tan(\alpha+\beta) = \frac{\tan\alpha + \tan\beta}{1 - \tan\alpha . \tan\beta}$$

$$= \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \times \frac{1}{3}} = \frac{\frac{3+2}{6}}{1 - \frac{1}{6}} = \frac{\frac{5}{6}}{\frac{5}{6}} = 1$$

tan (α + β) = 1 = tan 45°
α + β = 45°

92. If $\cot\theta = \frac{a}{b}$, then find the value of $\frac{\cos\theta - \sin\theta}{\cos\theta + \sin\theta}$

(a) b/a (b) (a – b)/(a + b)
(c) a/b (d) a^2/b^2

RRB JE - 25/05/2019 (Shift-I)

Ans : (b)

$$\frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}=\frac{\text{Sin}\theta\left(\frac{\cos\theta}{\sin\theta}-1\right)}{\sin\theta\left(\frac{\cos\theta}{\sin\theta}+1\right)}$$

$$=\frac{\cot\theta-1}{\cot\theta+1}\qquad\left[\because\cot\theta=\frac{a}{b}\right]$$

$$=\frac{\frac{a}{b}-1}{\frac{a}{b}+1}=\frac{a-b}{a+b}$$

93. If $\text{cosec}\,\alpha=\sqrt{2}$, then find the value of $\frac{2\sin^2\alpha+3\cos^2\alpha}{\text{cosec}^2\alpha+\cot^2\alpha}$

(a) $\frac{5}{12}$ (b) $\frac{5}{3}$

(c) $\frac{5}{2}$ (d) $\frac{5}{6}$

RRB JE - 01/06/2019 (Shift-III)

Ans. (d) $\text{cosec}\alpha=\sqrt{2}=\text{cosec}45°$

$\therefore\quad \alpha=45^0$

$$\frac{2\sin^2\alpha+3\cos^2\alpha}{\text{cosec}^2\alpha+\cot^2\alpha}=\frac{2\sin^2 45°+3\cos^2 45°}{\text{cosec}^2 45°+\cot^2 45°}$$

$$=\frac{2\times\left(\frac{1}{\sqrt{2}}\right)^2+3\times\left(\frac{1}{\sqrt{2}}\right)^2}{\left(\sqrt{2}\right)^2+1^2}$$

$$=\frac{2\times\frac{1}{2}+3\times\frac{1}{2}}{2+1}=\frac{1+\frac{3}{2}}{3}=\frac{\frac{5}{2}}{3}=\frac{5}{2}\times\frac{1}{3}=\frac{5}{6}$$

94. If $\sin\theta=\frac{3}{4}$ then What is the value of $16\cos^2\theta+\tan^2\theta$

(a) $\frac{58}{7}$ (b) $\frac{60}{7}$

(c) $\frac{55}{7}$ (d) $\frac{62}{7}$

RRB RPF SI – 10/01/2019 (Shift-I)

Ans : (a) $\sin\theta=\frac{3}{4}$ (given)

$$16\cos^2\theta+\tan^2\theta=16\cos^2\theta+\frac{\sin^2\theta}{\cos^2\theta}$$

$$=16\left(1-\sin^2\theta\right)+\frac{\sin^2\theta}{\left(1-\sin^2\theta\right)}$$

$$=16\times\left(1-\frac{9}{16}\right)+\frac{\frac{9}{16}}{\left(1-\frac{9}{16}\right)}$$

$$=16\times\left(\frac{7}{16}\right)+\frac{\frac{9}{16}}{\left(\frac{7}{16}\right)}=7+\frac{9}{7}=\frac{58}{7}$$

95. If $\tan\theta=\frac{4}{3}$, then the value of $\sin\theta+\cos\theta$?

(a) $\frac{6}{5}$ (b) $\frac{7}{5}$

(c) 1 (d) $\frac{4}{5}$

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (b) $\tan\theta=\frac{4}{3}$ then $\sin\theta+\cos\theta=?$

$$\tan\theta=\frac{\text{Perpendicular}}{\text{base}}$$

Perpendicular = 4

base = 3

Then hypotenuse $=\sqrt{(4)^2+(3)^2}=\sqrt{25}$

hypotenuse = 5

$$\sin\theta=\frac{\text{Perpendicular}}{\text{hypotenuse}}=\frac{4}{5}$$

$$\cos\theta=\frac{\text{base}}{\text{hypotenuse}}=\frac{3}{5}$$

then $\sin\theta+\cos\theta=\frac{4}{5}+\frac{3}{5}=\frac{7}{5}$

96. If 4 cotθ = 5, then find the value of $\frac{(5\sin\theta+3\cos\theta)}{(5\sin\theta-3\cos\theta)}$

(a) 3 (b) 9

(c) 7 (d) 4

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (c) $4\cot\theta=5\Rightarrow\cot\theta=\frac{5}{4}$

$$\frac{5\sin\theta+3\cos\theta}{5\sin\theta-3\cos\theta}=\frac{\sin\theta\left(5+3\frac{\cos\theta}{\sin\theta}\right)}{\sin\theta\left(5-3\frac{\cos\theta}{\sin\theta}\right)}$$

$$=\frac{5+3\cot\theta}{5-3\cot\theta}=\frac{5+3\times\frac{5}{4}}{5-3\times\frac{5}{4}}=\frac{5+\frac{15}{4}}{5-\frac{15}{4}}=\frac{35}{5}=7$$

97. If $\tan\theta=\frac{5}{6}$, then what is the value of $\frac{12\sin\theta-5\cos\theta}{12\sin\theta+5\cos\theta}$?

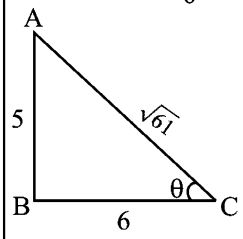
(a) $\frac{2}{3}$ (b) $\frac{1}{3}$

(c) $\frac{3}{4}$ (d) $\frac{1}{4}$

RRB Group-D – 19/09/2018 (Shift-III)

Ans. (b) :

$\tan\theta = \frac{5}{6}$ then $\frac{12\sin\theta - 5\cos\theta}{12\sin\theta + 5\cos\theta} = ?$



From Pythagoras theorem,
$AC^2 = AB^2 + BC^2$
$AC^2 = 5^2 + 6^2$
$AC^2 = 61$
$AC = \sqrt{61}$

$\sin\theta = \frac{5}{\sqrt{61}}$,

Putting the $\cos\theta$ $\frac{6}{\sqrt{16}}$ = value in equation

$$\frac{12\sin\theta - 5\cos\theta}{12\sin\theta + 5\cos\theta}$$

$$\Rightarrow \frac{12\times\frac{5}{\sqrt{61}} - 5\times\frac{6}{\sqrt{61}}}{12\times\frac{5}{\sqrt{61}} + 5\times\frac{6}{\sqrt{61}}}$$

$$\Rightarrow \frac{\frac{60-30}{\sqrt{61}}}{\frac{60+30}{\sqrt{61}}}$$

$$\Rightarrow \frac{\frac{30}{\sqrt{61}}}{\frac{90}{\sqrt{61}}} = \frac{30}{\sqrt{61}}\times\frac{\sqrt{61}}{90} = \frac{1}{3}$$

98. Find the value of $\sin\alpha\cos\beta - \cos\alpha\sin\beta$. If α and β complementary angle and $\operatorname{cosec}\alpha = 5/3$

(a) $\frac{-25}{7}$ (b) $\frac{-7}{25}$

(c) $\frac{7}{25}$ (d) $\frac{25}{7}$

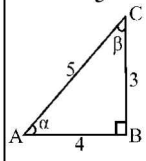
RRB Group-D – 30/10/2018 (Shift-III)

Ans. (b) :

$\operatorname{cosec}\alpha = \frac{5}{3}$

$\frac{1}{\sin\alpha} = \frac{5}{3}$

$\sin\alpha = \frac{3}{5}$



$$\sin\alpha = \frac{\text{Perpendicular}}{\text{hypotenuse}} = \frac{3}{5}$$

$$\cos\alpha = \frac{\text{base}}{\text{hypotenuse}} = \frac{4}{5}$$

$\because \sin\beta = \frac{4}{5}$

$\cos\beta = \frac{3}{5}$

$\sin\alpha\cos\beta - \cos\alpha\sin\beta$

$\frac{3}{5}\times\frac{3}{5} - \frac{4}{5}\times\frac{4}{5}$

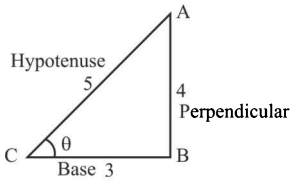
$\frac{9}{25} - \frac{16}{25} = \frac{-7}{25}$

99. If $\tan\theta = \frac{4}{3}$, then find the value of $\sin\theta + \cos\theta$. ($\sin\theta$ and $\cos\theta$ both are positive).

(a) 1 (b) 7

(c) $2\frac{1}{3}$ (d) $1\frac{2}{5}$

Ans. (d) :



$\tan\theta = \frac{4}{3}$

$AC^2 = AB^2 + BC^2$

$AC^2 = (4)^2 + (3)^2$

$AC^2 = \sqrt{16+9}$

$AC^2 = \sqrt{25}$

$AC = 5$

then, $\sin\theta = \frac{4}{5}, \cos\theta = \frac{3}{5}$

$\sin\theta + \cos\theta$

$= \frac{4}{5} + \frac{3}{5} = 1\frac{2}{5}$

100. If $\tan A = \frac{3}{4}$, then

$\{1/2\} + \{(1+\cos A)(1-\cos A)/(1+\sin A)(1-\sin A)\} - 1 = ?$

(a) $\frac{1}{16}$ (b) $\frac{12}{25}$

(c) $\frac{-9}{25}$ (d) $\frac{-1}{9}$

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (a) : If $\tan A = \frac{3}{4}$

$$\left\{\frac{1}{2}\right\} + \left\{\frac{(1+\cos A)(1-\cos A)}{(1+\sin A)(1-\sin A)}\right\} - 1 = ?$$

$$= \frac{1}{2} + \left\{\frac{(1-\cos^2 A)}{1-\sin^2 A}\right\} - 1 \quad = \frac{1}{2} + \left\{\frac{\sin^2 A}{\cos^2 A}\right\} - 1$$

$$= \frac{1}{2} + \tan^2 A - 1 \quad = \frac{1}{2} + \left(\frac{3}{4}\right)^2 - 1$$

$$= \frac{1}{2} + \frac{9}{16} - 1 = \frac{8+9-16}{16} \quad = \frac{17-16}{16}$$

$$\boxed{? = \frac{1}{16}}$$

101. If $\text{Tan A} = \frac{15}{8}$ and $\text{Tan B} = \frac{7}{24}$, then Tan(A - B) = ?

(a) $\frac{304}{297}$ (b) $\frac{304}{425}$

(c) $\frac{416}{87}$ (d) $\frac{87}{416}$

RRB NTPC 11.04.2016 Shift : 2

Ans : (a) $\text{Tan A} = \frac{15}{8}, \text{Tan B} = \frac{7}{24}$

$$\text{Tan}(A-B) = \frac{\text{Tan A} - \text{Tan B}}{1 + \text{Tan A Tan B}} = \frac{\frac{15}{8} - \frac{7}{24}}{1 + \frac{15}{8} \times \frac{7}{24}}$$

$$= \frac{\frac{45-7}{24}}{\frac{192+105}{8 \times 24}} = \frac{38 \times 8}{297} = \frac{304}{297}$$

102. If sin x = $\frac{4}{5}$ then cosec x + cot x =

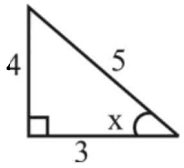
(a) 31/12 (b) 35/12

(c) 2 (d) 1/2

RRB NTPC 17.01.2017 Shift-1

Ans : (c) If $\sin x = \frac{4}{5}$ then $\cos x = \frac{3}{5}$

then cosec x + cot x



$$\frac{1}{\sin x} + \frac{\cos x}{\sin x}$$

$$\frac{1}{\frac{4}{5}} + \frac{3/5}{\frac{4}{5}}$$

$$\frac{5}{4} + \frac{3}{4} = \frac{8}{4} = 2$$

$$\boxed{\text{cosec}x + \cot x = 2}$$

103. If tanθ = $\frac{1}{\sqrt{5}}$, then $\text{cosec}^2\theta$ = ?

(a) 5 (b) $\sqrt{5}$

(c) $\sqrt{3}$ (d) 6

RRB NTPC 17.01.2017 Shift-2

Ans : (d) $\tan\theta = \frac{1}{\sqrt{5}}$

$\Rightarrow \cot\theta = \sqrt{5}$

$\text{cosec}^2\theta = 1 + \cot^2\theta$

$= 1 + \left(\sqrt{5}\right)^2 = 1 + 5 = 6$

104. If sinθ = 5/13, find cosθ.

(a) 8/13 (b) 12/13

(c) 23/13 (d) 1

RRB NTPC 04.04.2016 Shift : 1

Ans : (b) $\sin\theta = \frac{5}{13}$

$\sin^2\theta = \frac{25}{169}$

$\cos\theta = \sqrt{1 - \sin^2\theta} = \sqrt{1 - \frac{25}{169}} = \sqrt{\frac{144}{169}} = \frac{12}{13}$

105. Find the value of (3sin θ – 2 cosθ) ÷ (2 sinθ + 3 coθs). If 5 tanθ = 4.

(a) 6/23 (b) 2/23

(c) 4/23 (d) 5/23

RRB NTPC 31.03.2016 Shift : 3

Ans : (b) Given- $5\tan\theta = 4 \Rightarrow \tan\theta = \frac{4}{5}$

$$\frac{3\sin\theta - 2\cos\theta}{2\sin\theta + 3\cos\theta}$$

$$= \frac{3\left(\frac{\sin\theta}{\cos\theta}\right) - 2\left(\frac{\cos\theta}{\cos\theta}\right)}{2\left(\frac{\sin\theta}{\cos\theta}\right) + 3\left(\frac{\cos\theta}{\cos\theta}\right)}$$

(divided by cosθ in numerator and denominator)

$$= \frac{3\tan\theta - 2 \times 1}{2\tan\theta + 3} = \frac{3 \times \frac{4}{5} - 2}{2 \times \frac{4}{5} + 3} = \frac{\frac{12-10}{5}}{\frac{8+15}{5}} = \frac{2}{23}$$

106. Find the value of cos2θ. If $\sqrt{3}$tanθ = 1.

(a) 1/2 (b) $1/\sqrt{3}$

(c) 1/3 (d) 1

RRB NTPC 12.04.2016 Shift : 1

Ans : (a) From the question,

$\sqrt{3}\tan\theta = 1$

$\Rightarrow \tan\theta = \frac{1}{\sqrt{3}} = \tan 30^0$

$\therefore \theta = 30^0$

$\therefore \cos 2\theta = \cos 2 \times 30^0$

$= \cos 60^0 = \frac{1}{2}$

107. If $\text{Sin x} = \frac{4}{5}$, then $\frac{\text{Sec x}}{\text{Sin x}}$ = ?

(a) $\frac{23}{12}$ (b) $\frac{25}{4}$

(c) $\frac{4}{5}$ (d) $\frac{25}{12}$

RRB NTPC 11.04.2016 Shift : 2

Ans : (d) $\text{Sin x} = \frac{4}{5}$

$\sin x = \frac{\text{Perpendicualr}}{\text{hypotenuse}}$

$(\text{base})^2 = (\text{hypotenuse})^2 - (\text{Perpendicular})^2$

base = $\sqrt{(5)^2-(4)^2}$

base = $\sqrt{25-16}$

base = $\sqrt{9}$

base = 3

then $\sec x = \frac{\text{hypotenuse}}{\text{base}} = \frac{5}{3}$

$$\frac{\sec x}{\sin x} = \frac{\frac{5}{3}}{\frac{4}{5}} = \frac{5}{3}\times\frac{5}{4} = \frac{25}{12}$$

108. If $\sin x = \frac{4}{5}$, then tanx + cosx = ?

(a) $\frac{31}{15}$ (b) $\frac{29}{12}$

(c) $\frac{15}{29}$ (d) $\frac{29}{15}$

RRB NTPC 07.04.2016 Shift : 2

Ans : (d)

$\sin x = \frac{4}{5} = \frac{\text{Perpendicular}}{\text{hypotenuse}}$

$\therefore$ base = $\sqrt{\text{hypotenuse}^2 - \text{Perpendicular}}$

$= \sqrt{(5)^2-(4)^2} = \sqrt{25-16} = \sqrt{9} = 3$

$\therefore$ from the question,

tan x + cos x

$= \frac{\text{Perpendicular}}{\text{base}} + \frac{\text{base}}{\text{hypotenuse}}$

$= \frac{4}{3} + \frac{3}{5} = \frac{20+9}{15} = \frac{29}{15}$

109. If $\tan A = \frac{15}{8}$ and $\tan B = \frac{7}{24}$, then cosec (A–B) = ?

(a) $\frac{425}{304}$ (b) $\frac{425}{416}$

(c) $\frac{425}{87}$ (d) $\frac{425}{297}$

RRB NTPC 07.04.2016 Shift : 2

Ans : (a) $\tan A = \frac{15}{8}$ and $\tan B = \frac{7}{24}$

$\frac{\text{Perpendicular}}{\text{base}} = \frac{15}{8}$ $\frac{\text{Perpendicular}}{\text{base}} = \frac{7}{24}$

hypotenuse = $\sqrt{(15)^2+(8)^2}$ hypotenuse = $\sqrt{(24)^2+(7)^2}$

$= \sqrt{225+64}$ $= \sqrt{576+49}$

$= \sqrt{289}$ $= \sqrt{625}$

$= 17$ $= 25$

$\therefore$ sin (A–B) = sinA . cos B – cosA . sinB

$\text{cosec}(A-B) = \frac{1}{\sin(A-B)}$

$= \frac{1}{\sin A.\cos B - \cos A.\sin B}$

$= \frac{1}{\frac{15}{17}\times\frac{24}{25} - \frac{8}{17}\times\frac{7}{25}}$ $= \frac{1}{\frac{360}{425} - \frac{56}{425}}$

$= \frac{425}{360-56} = \frac{425}{304}$

$\therefore \text{cosec}(A-B) = \frac{425}{304}$

110. If 2cosθ = √3, then cosθ × tanθ = ?

(a) 1 (b) √3/3

(c) √3/2 (d) 1/2

RRB NTPC 06.04.2016 Shift : 2

Ans : (d) $2\cos\theta = \sqrt{3}$ $\cos\theta = \frac{\sqrt{3}}{2}$

$\cos\theta = \cos 30^0$

$\theta = 30^0$

$\cos\theta \times \tan\theta = \cos 30 \times \tan 30 = \frac{\sqrt{3}}{2}\times\frac{1}{\sqrt{3}} = \frac{1}{2}$

111. If sin x = 4/5 then sec x + tan x = ?

(a) 37/20 (b) 31/12

(c) 3 (d) 1/3

RRB NTPC 19.01.2017 Shift : 3

Ans : (c) $\sin x = \frac{4}{5}$

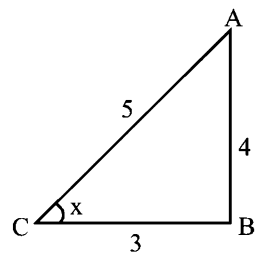
from triangle ABC,

$BC = \sqrt{AC^2 - AB^2}$

$= \sqrt{(5)^2-(4)^2}$

$= \sqrt{25-16} = \sqrt{9}$

BC = 3 unit



$\sec x + \tan x = \frac{5}{3} + \frac{4}{3}$

$\Rightarrow \frac{9}{3} = 3$

$\boxed{\sec x + \tan x = 3}$

112. If A + B = 90° and $\cos B = \frac{1}{3}$, then the value of sin A

(a) 1/2 (b) 1/4

(c) 1/3 (d) 2/3

RRB NTPC 27.04.2016 Shift : 1

Ans : (c) A + B = 90°

B = 90° – A

taking cos both side

cosB = cos (90°–A)

$\because \cos B = \frac{1}{3}$

$\cos(90^\circ - A) = \frac{1}{3}$

$\sin A = \frac{1}{3}$

113. If $\sin A = \frac{4}{5}$ and $\sin B = \frac{5}{13}$ then cot (A–B)=?

(a) $\frac{56}{33}$ (b) $\frac{65}{33}$

(c) $\frac{33}{56}$ (d) $\frac{16}{33}$

RRB NTPC 30.04.2016 Shift : 3

Ans : (a) $\sin A = \frac{4}{5}$

$$\sin A = \frac{\text{Perpendicular}}{\text{hypotenuse}} = \frac{4}{5}$$

$$\text{base} = \sqrt{(\text{Hypotenuse})^2 - (\text{Perpendicular})^2}$$

$$\text{base} = \sqrt{(5)^2 - (4)^2}$$

$$\text{base} = \sqrt{25-16}$$

$$\text{base} = \sqrt{9}$$

$$\text{base} = 3$$

$$\cot A = \frac{\text{base}}{\text{perpendicular}}$$

$$\cot A = \frac{3}{4}$$

Likwise

$$\sin B = \frac{5}{13} = \frac{\text{Perpendicular}}{\text{hypotenuse}}$$

then base = 12

$$\therefore \quad \cot B = \frac{12}{5}$$

$$\cot(A-B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$$

$$= \frac{\frac{3}{4}\times\frac{12}{5}+1}{\frac{12}{5}-\frac{3}{4}} = \frac{\frac{36+20}{20}}{\frac{48-15}{20}} = \frac{56}{33}$$

114. If $\tan\theta = \frac{7}{24}$ then find the value of p such that $\frac{\tan\theta - \sec\theta}{\sin\theta} = \frac{-p}{28}$

(a) 25 (b) 75

(c) 50 (d) 100

RRB ALP & Tec. (09-08-18 Shift-III)

Ans : (b) given- $\tan\theta = \frac{7}{24} = \frac{\text{Perpendicular}}{\text{base}}$

$$\therefore \text{hypotenuse} = \sqrt{(\text{base})^2 + (\text{Perpendicular})^2}$$

$$= \sqrt{24^2 + 7^2} = \sqrt{625} = 25$$

$\therefore$ from the question

$$\frac{\tan\theta - \sec\theta}{\sin\theta} = \frac{-P}{28}$$

$$\Rightarrow \frac{\frac{7}{24} - \frac{25}{24}}{\frac{7}{25}} = \frac{-P}{28}$$

$$\Rightarrow \frac{-18}{24} \times \frac{25}{7} = \frac{-P}{28}$$

$$P = \frac{18 \times 25 \times 28}{24 \times 7} = 75$$

Type - 6

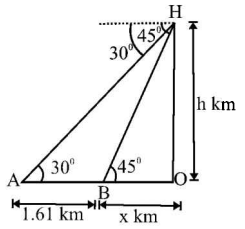
115. The angle of depression of two points in the same direction on the road from an aeroplane over a flat road are 45° and 30° respectively. The distance between the two points is one mile (1.61 km). Approximate distance of aeroplane from land in kilometers?

(a) 1.8 (b) 2.2

(c) 2.4 (d) 2.0

RRB RPF SI – 11/01/2019 (Shift-II)

Ans. (b) : Suppose distance of ship from the land = h km



in Δ BOH,

$$\tan 45° = \frac{h}{x}$$

$$h = x \quad \text{.....(i)}$$

in Δ AOH,

$$\tan 30° = \frac{h}{1.61 + x}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{1.61 + x}$$

$$\sqrt{3}h = 1.61 + x$$

$$\sqrt{3}h = 1.61 + h \quad \{\because \text{ from the equation (i)}\}$$

$$(1.732 - 1)h = 1.61$$

$$h = \frac{1.61}{0.732}$$

$$= 2.19 \approx 2.2 \text{ km}$$

$$h = 2.2 \text{ km}$$

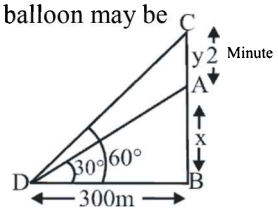
116. The angle of elevation of a hot air balloon going upward in vertical direction when viewed from a distance of 300 m from the flight point at 10:00 am, changing from 30° to 60° at 10:02 am. Find the speed of the balloon in the vertical direction.

(a) 2 m/sec. (b) 2.18 m/sec.

(c) 3.4 m/sec. (d) 2.9 m/sec.

RRB JE - 22/05/2019 (Shift-III)

Ans : (d) Suppose position of balloon may be A to C in 2 minutes.



in ΔABD,

$$\tan 30° = \frac{x}{300}$$

$$\frac{1}{\sqrt{3}} = \frac{x}{300}$$

$$x = \frac{300}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 100\sqrt{3} \text{ m}$$

in ΔCBD,

$$\tan 60° = \frac{x+y}{300}$$

$$\sqrt{3} = \frac{x+y}{300}$$

$$x + y = 300\sqrt{3}$$

$$100\sqrt{3} + y = 300\sqrt{3}$$

$$y = 200\sqrt{3}$$

$\therefore$ Speed of balloons $= \frac{y}{t} = \frac{200\sqrt{3}}{2 \times 60} = \frac{5}{3}\sqrt{3}$

$= \frac{5 \times 1.732}{3} = 5 \times 0.577 = 2.885 = 2.9$ m/sec

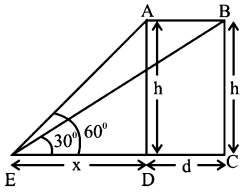
117. An aeroplane is flying at a steady altitude 'h'. At 10:00 am, it appears at an elevation angle of 30°. After 1 minute, it appears at an elevation angle of 60°. If the speed of the aeroplane is 960 km/h, find the value of 'h'.

(a) 15 km. (b) 13.86 km.
(c) 20 km. (d) 12.46 km.

RRB JE - 26/05/2019 (Shift-I)

Ans : (b)

$\therefore$ Speed = distance/time



$$960 = \frac{CD}{1 \text{minute}}$$

$$\frac{960 \times 1}{60} = d \Rightarrow d = 16\text{km}$$

from ΔAED–

$$\tan 60° = \frac{h}{x} \Rightarrow x\sqrt{3} = h.....(i)$$

from ΔBEC–

$$\tan 30^0 = \frac{h}{x+d} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{16+x}.....(ii)$$

from equation (i) and (ii)–

$$\frac{1}{\sqrt{3}} = \frac{x\sqrt{3}}{16+x}$$

$16 + x = 3x$

$2x = 16$

$x = 8$ km

$\therefore h = x\sqrt{3} = 8\sqrt{3}$

$h = 1.732 \times 8 = 13.86$ km

118. The angle of elevation of a steep ladder supported by a wall is 45° and its lower end is 10 meters from the wall. Find height of the ladder.

(a) $10\sqrt{2}$ meter (b) 13 meter
(c) 15 meter (d) 16 meter

RRB RPF Constable – 19/01/2019 (Shift-II)

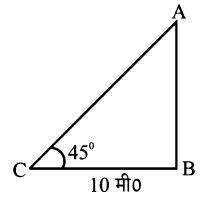
Ans : (a)

As per figure–

height of ladder = AC

in Δ ABC –

$\cos 45^0 = BC/AC$



$$\frac{1}{\sqrt{2}} = \frac{10}{AC}$$

$AC = 10\sqrt{2}$ meter

119. The elevation angles formed when looking at the top of the tree from two points located 'x' meters and 'y' meters from both the foot of the tree are α, β respectively. If α + β = 90°, then find the height of tree.

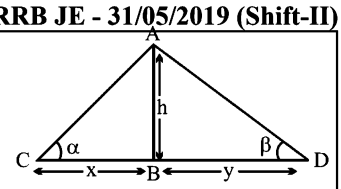
(a) $\frac{(x+y)^2}{2}$ (b) $\sqrt{xy}$

(c) $x\cos a + y\cos B$ (d) $(x-y)^2$

RRB JE - 31/05/2019 (Shift-II)

Ans : (b)

Suppose height of is h.



$\alpha + \beta = 90°$ ----(i)

in ΔABC,

$\tan \alpha = \frac{h}{x}$ ----(ii)

in ΔABD,

$\tan \beta = \frac{h}{y}$

$\tan (90-\alpha) = \frac{h}{y}$ {from equation (i) }

$\cot \alpha = \frac{h}{y}$ ----(iii)

from equation (ii) and (iii)-

$$\tan \alpha \cot \alpha = \frac{h}{x} \times \frac{h}{y}$$

$$1 = \frac{h^2}{xy}$$

$$h = \sqrt{xy}$$

120. The angle of depression of the foot of a building from the top of a tower of $32\sqrt{3}$ meters height is 60°. How far is the building from the tower?

(a) 32 meter (b) $16\sqrt{3}$ meter
(c) $32\sqrt{3}$ meter (d) 16 meter

RRB RPF SI – 06/01/2019 (Shift-II)

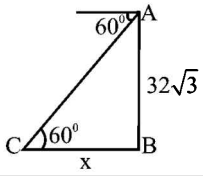
Ans. (a) : height of tower AB = $32\sqrt{3}$ m

Let distance of tower to building is x m.

Then $\tan 60^0 = \frac{AB}{CB}$

$\sqrt{3} = \frac{32\sqrt{3}}{x}$

x = 32 m



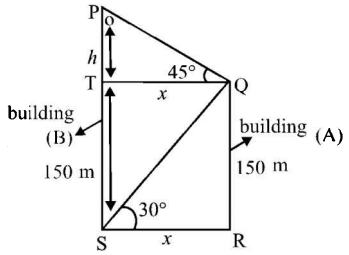
121. A 45° elevation angle is formed from the top of a 150 meter high building A to the top of building B. A depression angle of 30° is formed at the bottom of building B. What is the height of building B?

(a) $150\left(1+\sqrt{3}\right)$m (b) 250 m

(c) $150\sqrt{3}$m (d) $\frac{450}{\sqrt{3}}$m

RRB RPF Constable – 20/01/2019 (Shift-III)

Ans : (a)



in ΔQPT -

$\tan 45° = \frac{h}{x}$

$1 = \frac{h}{x}$

$h = x$ _____ (i)

in ΔSRQ -

$\tan 30° = \frac{150}{x}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{150}{x}$

$x = 150\sqrt{3}$

$h = 150\sqrt{3}$

∴ Total height of building (B) = 150 + h

$= 150 + 150\sqrt{3}$

$= 150\ (1+\sqrt{3})$m

122. The angle of elevation of the top of a large building from a 18 meter high tower is 45° and the angle of depression of bottom of the same building is 60°. What is the height of building (in meters)?

(a) $6\left(3+\frac{\sqrt{3}}{2}\right)$ (b) $6\left(3+\sqrt{3}\right)$

(c) $18+\sqrt{2}$ (d) $12+6\sqrt{3}$

RRB Group-D – 20/09/2018 (Shift-II)

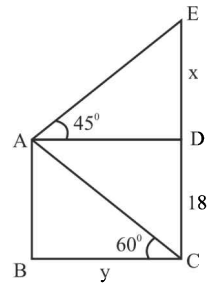
Ans : (b)

in ΔABC

$\tan 60^0 = \frac{18}{y}$

$\sqrt{3} = \frac{18}{y}$

∴ $y = \frac{18}{\sqrt{3}} = 6\sqrt{3}$



in ΔADE

$\tan 45° = \frac{x}{y} = \frac{x}{6\sqrt{3}}$

$1 = \frac{x}{6\sqrt{3}}$

$x = 6\sqrt{3}$

So x = y

then height of building = 18 + x

$CE = 18 + 6\sqrt{3}$

$CE = 6\left(3+\sqrt{3}\right)$ m

123. A 13 meter long ladder is placed along the wall and the lower part of the ladder is 6.5 meter from the wall. What is the angle of elevation of the ladder tilted along the wall?

(a) 45^0 (b) 60^0

(c) 105^0 (d) 30^0

RRB Group-D – 22/09/2018 (Shift-I)

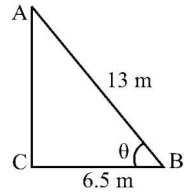
Ans : (b) Given

length of ladder AB = 13m

distance from wall = 6.5 m

Let angle of height is θ

in Δ ACB



$\cos\theta = \frac{\text{base}}{\text{hypotenuse}}$

$\Rightarrow \cos\theta = \frac{BC}{AB} = \frac{6.5}{13} = \frac{1}{2}$

$\Rightarrow \cos\theta = \cos\ 60°$

$\theta = 60°$ $\left[\because \cos 60° = \frac{1}{2}\right]$

So angle of height θ = 60°

124. From the top of the stage the elevation angle of the top of the tower at a distance of $50\sqrt{3}$ meter is 30°. If the height of the tower is 60 meter; what will be the height of the stage?

(a) $20\sqrt{3}$ m (b) 10 m

(c) 40 m (d) $45\sqrt{3}$ m

RRB Group-D – 28/09/2018 (Shift-III)

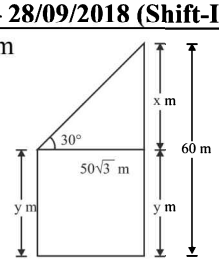
Ans : (b) Let height of stage = y m

As per question—

$\tan 30° = \frac{x}{50\sqrt{3}}$

$\frac{1}{\sqrt{3}} = \frac{x}{50\sqrt{3}}$

$x = 50$ m



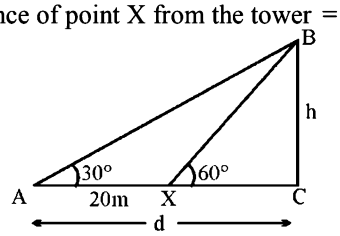
$\because x + y = 60$
$y = 60 - x$
$y = 60 - 50$
$y = 10$ m
so height of stage = 10 m

125. The angle of elevation of a point A from the top of a tower to the ground is 30°. When moving to the point X at 20 meter towards the foot of the tower, the elevation angle changes to 60°. What is the distance of the tower from point A?

(a) $5\sqrt{11}$ m (b) 30 m
(c) 5 m (d) 16m

RRB Group-D – 28/09/2018 (Shift-III)

Ans : (b) Let height of tower CB = h m
and distance of point A from the tower = d m
∴ distance of point X from the tower = (d – 20)m



From the figure, $\tan 30° = \frac{h}{d}$

$\frac{1}{\sqrt{3}} = \frac{h}{d}$

$h = \frac{d}{\sqrt{3}}$

and

$\tan 60° = \frac{h}{(d-20)} \Rightarrow \sqrt{3} = \frac{h}{(d-20)}$

Putting the value of 'h'

$\sqrt{3} = \frac{d}{\sqrt{3}(d-20)}$

$3(d - 20) = d$
$3d - d = 60$
$2d = 60$
$d = 30$

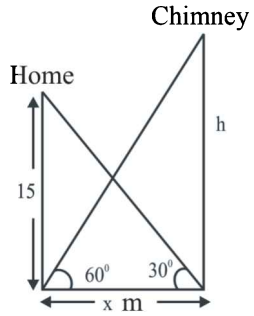
So distance of point A from the base = 30 m.

126. A 15 meter tall house situated on the banks of a river opposite to a chimney situated on the other bank of the river. The angle of depression of the bottom of chimney from the roof of the house is 30° and the elevation angle of the top of the chimney from the bottom of the house becomes 60°. Find the approximate width of the river and the height of the chimney.

(a) 24 m and 47 m
(b) 20 m and 25 m
(c) 26 m and 45 m
(d) 25 m and 41 m

RRB Paramedical Exam – 21/07/2018 (Shift-II)

Ans : (c) ΔABC in—
Let breadth of river = x m



$\tan 30° = \frac{15}{x}$

$\frac{1}{\sqrt{3}} = \frac{15}{x}$

$x = 15\sqrt{3}$ _____(i)

$\tan 60° = \frac{h}{x}$

$\sqrt{3} = \frac{h}{x}$

$h = \sqrt{3} \times x$ _____(ii)

Putting the value of equation (i), in equation (ii)

$h = \sqrt{3} \times 15\sqrt{3} = 45$m

breadth of river $(x) = 15 \times 1.732 = 25.98 \simeq 26$m

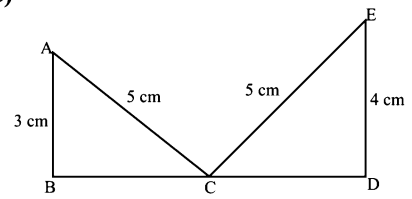
height of chimney $(h) = 45$m

127. A 5 meter long staircase with a stable base on the road can reach two windows 4 meter and 3 meter high on the other side of the road. What is the width of the road?

(a) 6.2 m (b) 6 m
(c) 7 m (d) 5.5 m

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (c)



$BD = \sqrt{BC^2} + \sqrt{CD^2}$

$= \sqrt{5^2 - 3^2} + \sqrt{5^2 - 4^2}$

$= \sqrt{4^2} + \sqrt{3^2} = 4 + 3 = 7$ m

So breadth of road = 4 + 3 = 7m

128. From the top of a platform of 7 meter height, the elevation angle of a tower which is 47 meter in height is 30°. How far is the tower located from the platform.

(a) $45\sqrt{3}$ m (b) 40 m
(c) $40\sqrt{3}$ m (d) $15\sqrt{3}$ m

RRB Group-D – 18/09/2018 (Shift-II)

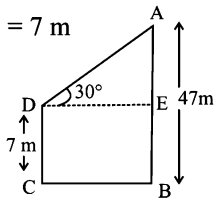
Ans. (c) :

$\because$ AB = height of tower = 47 m

CD = BE = height of platform = 7 m

$\therefore$ AE = 47 – 7 = 40 m

from ΔADE ,



$$\tan 30° = \frac{AE}{ED} = \frac{40}{DE}$$

$\because$ DE = BC =

(distance between platform to tower)

$$\therefore \quad \frac{1}{\sqrt{3}} = \frac{40}{DE}$$

$$DE = 40\sqrt{3}\text{ m or } BC = 40\sqrt{3}\text{ m}$$

129. The elevation angle of the top was 40° from the initial position of a woman standing on the ground floor with a 43.5 tall tower. She moves in a straight line from the position of the tower in such way that the angle of the alevation of the tower from its final position becomes 30° then what will be the changed distance.

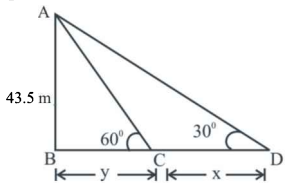
(a) $\frac{29}{3}\sqrt{3}$ meter (b) $29\sqrt{3}$ meter

(c) $\frac{29}{2}\sqrt{3}$ meter (d) 29 meter

RRB Group-D – 20/09/2018 (Shift-I)

Ans. (b) : Let primary distance of lady from the tower = y m. and distance after change place = x m.

(AB) height of tower = 43.5 m



in ΔABC,

$$\text{Tan}60° = \frac{AB}{BC}$$

$$\sqrt{3} = \frac{43.5}{y}$$

$$y = \frac{43.5}{\sqrt{3}}$$

$$\boxed{3y = 43.5\sqrt{3}}\text{.....(i)}$$

in ΔABD,

$$\text{Tan}30° = \frac{AB}{BC}$$

$$\frac{1}{\sqrt{3}} = \frac{43.5}{x+y}$$

$$\boxed{x + y = 43.5\sqrt{3}}\text{.....(ii)}$$

Putting the value $43.5\sqrt{3}$ in equation (ii)

$$x + y = 3y$$

$$x = 2y$$

$$y = \frac{x}{2}$$

Again putting the value of y in equation (i)

$$3 \times \frac{x}{2} = 43.5\sqrt{3}$$

$$x = \frac{43.5\sqrt{3} \times 2}{3}$$

$$x = \frac{87\sqrt{3}}{3}$$

$$= 29\sqrt{3} \text{ m}$$

130. The elevation angle of the top of a tower 78 feet high is 30° from the initial position of the person standing on the ground. He moves towards the tower in such a way that he finds its initial position and final position are all in the same straight line. The angle of elevation of the top of the tower from its final position is 60°. How far did he go from her initial position?

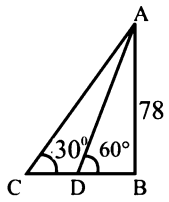
(a) 52 feet (b) $26\sqrt{3}$ feet

(c) $52\sqrt{3}$ feet (d) $26\frac{\sqrt{3}}{3}$ feet

RRB Group-D – 15/11/2018 (Shift-III)

Ans. : (c) Tower AB = 78 feet

in ΔABC,



$$\tan 30^0 = \frac{78}{(CD + BD)}$$

$CD + BD = 78\sqrt{3}$ feet ----(i)

in Δ ABD

$$\tan 60^0 = \frac{78}{BD}$$

$$BD = \frac{78}{\sqrt{3}} \text{------(ii)}$$

Putting the value of BD in equation (i)

$$CD = 78\sqrt{3} - \frac{78}{\sqrt{3}}$$

$$CD = \frac{156}{\sqrt{3}}$$

$$CD = \frac{156\sqrt{3}}{\sqrt{3} \times \sqrt{3}}$$

$$CD = 52\sqrt{3} \text{ feet}$$

So the travels a distance of $52\sqrt{3}$ feet from his initial to his final position.

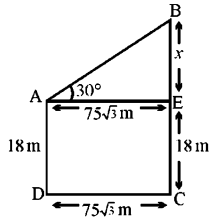
131. From the top of a platform 18 meter high, the elevation angle of the top of the tower is 30°. If the platform is located $75\sqrt{3}$ meter away from the tower, then what is the height of the tower?

(a) 93 m (b) $50\sqrt{3}$ m

(c) 75 m (d) $37.5\sqrt{3}$ m

RRB Group-D – 01/11/2018 (Shift-II)

Ans : (a) Drawing the diagram according to the statement.



in ΔABE,

$$\tan 30° = \frac{x}{75\sqrt{3}}$$

$$\frac{1}{\sqrt{3}} = \frac{x}{75\sqrt{3}}$$

$x = 75m$

So height of Tower = 18 + 75 = 93m

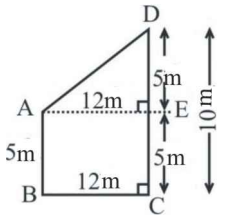
132. Two pillars of 5 meter and 10 meter stand straight on the ground. If the distance between their bottoms is 12 meter, find the distance between their tops.

(a) 11 meter (b) 12 meter

(c) 13 meter (d) 14 meter

RRB NTPC 04.04.2016 Shift : 3

Ans : (c)



Let distance between top of the pillars = AD m

$AE = BC = 120$ m

$DE = CD - CE = 10 - 5$

DE = 5 m

from pythagoras theorem in ΔAED

$AD^2 = AE^2 + DE^2 = (12)^2 + (5)^2 = 144 + 25$

$AD^2 = 169 \Rightarrow AD = 13$ m

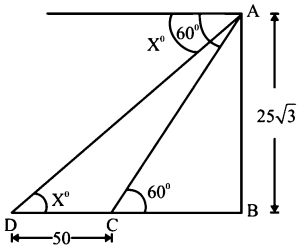
133. The shadow of a tower of $25\sqrt{3}$ height increases by 50 meters when the angle of depression by the sun is 60° to X°. Find the measure of X.

(a) 45^0 (b) 30^0

(c) 75^0 (d) 90^0

RRB NTPC 19.04.2016 Shift : 1

Ans : (b)



In ΔABC

$$\tan 60^0 = \frac{25\sqrt{3}}{BC}$$

$$\sqrt{3} = \frac{25\sqrt{3}}{BC}$$

BC = 25 …… (1)

again in ΔABD,

$$\tan x = \frac{25\sqrt{3}}{50 + BC}$$

$$\tan x = \frac{25\sqrt{3}}{50+25} = \frac{25\sqrt{3}}{75} = \frac{1}{\sqrt{3}}$$

$\tan x = \tan 30^0$

$x = 30^0$

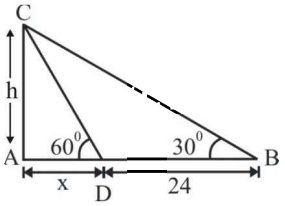
134. The length of the shadow of a pillar decreases by 24 meter. When the angle of elevation of the sun increases from 30° to 60°, the length of the pllar is.

(a) $10\sqrt{3}$ (b) $8\sqrt{3}$

(c) $16\sqrt{3}$ (d) $12\sqrt{3}$

RRB NTPC 18.04.2016 Shift : 2

Ans : (d)



$$\text{Tan } 30^0 = \frac{h}{x+24}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{x+24}$$

$\sqrt{3}h = x + 24$ .....(I)

$$\text{Tan } 60^0 = \frac{h}{x}$$

$$\sqrt{3} = \frac{h}{x}$$

$h = \sqrt{3}x$ .....(II)

Putting the value of h in equation (I)

$\sqrt{3} \times \sqrt{3}x = x + 24$

$3x = x + 24$

$2x = 24$

$x = 12$

$$\text{Tan } 60^0 = \frac{h}{12}$$

$$\sqrt{3} = \frac{h}{12}$$

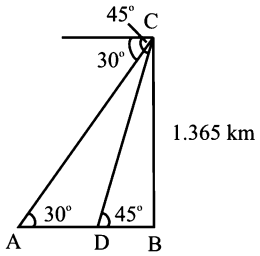
$h = 12\sqrt{3}$ m

135. The angle of depression of two stones in the same direction from an aeroplane vertically above a straight road is 30° and 45° respectively. If the plane is flying at an altitude of 1.365 km, then what is the distance between the two stones?

(a) 1 km (b) 2 km
(c) 3 km (d) 4 km

RRB NTPC 16.04.2016 Shift : 2

Ans : (a) ∵ from ΔDBC



$$\tan 45^\circ = \frac{1.365}{DB}$$

DB = 1.365 km

from Δ ABC,

$$\tan 30^\circ = \frac{1.365}{AB}$$

$AB = 1.365\sqrt{3}$

∵ AD = AB – DB

$AD = 1.365\sqrt{3} - 1.365 = 1.365\left(\sqrt{3}-1\right)$

$= 1.365 \times 0.73 = 0.99 \approx 1$ km

So distance between the stones is 1 km.

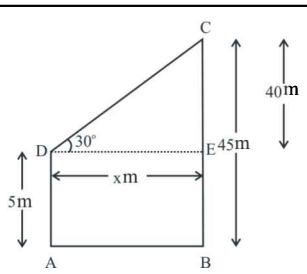
136. From the top of a platform of 5 m height, the angle of elevation of a tower was 30°. If the tower was 45 m high, how far was the platform from the tower.

(a) 40m/40 meter
(b) $40\sqrt{3}$ m\ $40\sqrt{3}$ meter
(c) $45\sqrt{3}$ m\ $45\sqrt{3}$ meter
(d) $15\sqrt{3}$ m\ $15\sqrt{3}$ meter

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (b)

from picture Δ DEC,



$$\tan 30^0 = \frac{CE}{DE}$$

$$\tan 30^0 = \frac{40}{x} \Rightarrow \frac{1}{\sqrt{3}} = \frac{40}{x}$$

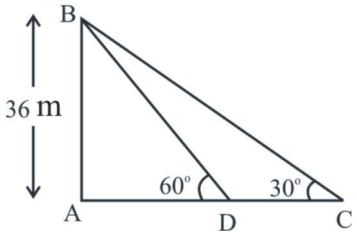
$x = 40\sqrt{3}$ m

137. The angle of elevation of the top of a 36 m long tower from the initial position of a woman on the ground was 60°. She walked away in a manner that the bottom of the tower, her initial position and the final position were all in the same straight line. The angle of elevation of the top of the tower from her final position was 30°. How much did she walk from her initial position?

(a) 24m/24 meter
(b) $36\sqrt{3}$ m/ $36\sqrt{3}$ meter
(c) $24\sqrt{3}$ m\ $24\sqrt{3}$ meter
(d) $12\sqrt{3}$ m\ $12\sqrt{3}$ meter

RRB ALP & Tec. (21-08-18 Shift-II)

Ans : (c)



From Δ ABC–

$$\tan 30^\circ = \frac{AB}{AC}$$

$$\tan 30^\circ = \frac{36}{AC}$$

$AC = 36 \times \sqrt{3}$

$AD + DC = 36 \times \sqrt{3}$ ..... (i)

From Δ ABD–

$$\tan 60^\circ = \frac{AB}{AD} = \frac{36}{AD}$$

$AD = \frac{36}{\sqrt{3}}$ ..... (ii)

from equation (i) and (ii)

$$\frac{36}{\sqrt{3}} + DC = 36\sqrt{3}$$

$$DC = 36\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right)$$

$$= 36 \times \frac{2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = 24\sqrt{3} \text{ m}$$

138. The angle of elevation of the top of a hill from the foot of the tower is 60° and the angle of elevation of the top of the tower from the foot of the hill is 30°. If the tower is 50m high, then what is the height of the hill?

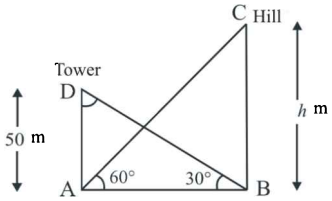
(a) 100m/100 meter
(b) 120m/120 meter
(c) 180m/180 meter
(d) 150m/150 meter

RRB ALP & Tec. (14-08-18 Shift-I)

Ans : (d) In ΔABD

$$\tan 30° = \frac{50}{AB} \Rightarrow \frac{1}{\sqrt{3}} = \frac{50}{AB}$$

$$\Rightarrow AB = 50\sqrt{3}\text{ m}$$



∴ In Δ ABC,

$$\tan 60° = \frac{h}{50\sqrt{3}}$$

$$\Rightarrow h = 50\sqrt{3} \times \sqrt{3}$$

$$\Rightarrow h = 150\text{ m}$$

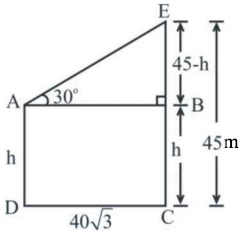
So height of hill is 150 meter.

139. From the top of a platform, the angle of elevation of a tower was 30°. The tower was 45 m high and the horizontal distance between the platform and the tower was $40\sqrt{3}$ m. What was the height of the platform?

(a) 40 m/40 meter
(b) 5 m /5 meter
(c) 45 m/ $45\sqrt{3}$ meter
(d) $20\sqrt{3}$ m/ $20\sqrt{3}$ meter

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (b) Given the height of tower EC = 45 meter



Elevation ∠EAB = 30°

Horizontal distance DC = $40\sqrt{3}$ meter = AB

Let height of platform is h meter.

∴ in ΔABE -

$$\tan 30° = \frac{EB}{AB}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{45-h}{40\sqrt{3}}$$

$$\Rightarrow 45 - h = 40$$

$$\Rightarrow h = 45 - 40$$

$$\Rightarrow h = 5\text{meter}$$

Type - 6

140. If 0°< θ ≤ 90°, then what is the value of θ, where $\cos^2\theta - 3\cos\theta + 2 = 2\sin^2\theta$

(a) 30° (b) 60°
(c) 90° (d) 45°

RRB RPF SI – 10/01/2019 (Shift-I)

Ans : (c) if 0°< θ ≤ 90°

then $\cos^2\theta - 3\cos\theta + 2 = 2\sin^2\theta$

$\cos^2\theta - 3\cos\theta + 2 = 2(1-\cos^2\theta)$

$\cos^2\theta - 3\cos\theta + 2 = 2 - 2\cos^2\theta$

$3\cos^2\theta = 3\cos\theta$

$3\cos^2\theta - 3\cos\theta = 0$

$\cos^2\theta - \cos\theta = 0$

$\cos\theta\,(\cos\theta - 1) = 0$

if $\cos\theta = 0° = \cos 90°$

then θ = 90°

if $\cos\theta = 1 = \cos 0°$

θ = 0°

141. If $3\sec^2 x - 2\tan^2 x = 6$ and $0^0 \le x \le 90^0$ then x = ?

(a) 60^0 (b) 45^0
(c) 30^0 (d) 90^0

RRB Group-D – 20/09/2018 (Shift-III)

Ans : (a) $3\sec^2 x - 2\tan^2 x = 6$

$3(1+\tan^2 x) - 2\tan^2 x = 6$ [$\sec^2 x = 1 + \tan^2 x$]

$3 + 3\tan^2 x - 2\tan^2 x = 6$

$\tan^2 x = 3$

$\tan x = \sqrt{3}$

$\tan x = \tan 60^0$

x = 60

142. If $8\sec^2 x - 7\tan^2 x = 11$ and $0^0 \le x \le 90^0$, then find the value of x

(a) 90^0 (b) 45^0
(c) 30^0 (d) 60^0

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (d) : If $8\sec^2 x - 7\tan^2 x = 11$ then x = ?

$\Rightarrow 8\sec^2 x - 7\tan^2 x = 11$

$\Rightarrow 8(1+\tan^2 x) - 7\tan^2 x = 11$

$\Rightarrow 8 + 8\tan^2 x - 7\tan^2 x = 11$

$\Rightarrow \tan^2 x = 11 - 8$

$\Rightarrow \tan^2 x = 3$

$\Rightarrow \tan x = \sqrt{3}$

$\Rightarrow \tan x = \tan 60^0$

$\boxed{x = 60^0}$